Multiagent Interval Consensus With Flocking Dynamics

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***Abstract*—In this article, we investigate the interval consensus for a network of agents with flocking dynamics, i.e., second-order multiagent systems, where each agent imposes an interval constraint on its preferred consensus values, with the aim of driving the agent into a favorable interval. Specifically, we work on two different frameworks of interval constraints, viz., the first one that the node states are constrained in their own constraint intervals and the second one that the node states are constrained in their neighbors' constraint intervals. For both of the frameworks, we provide a complete solution to the equilibrium seeking problem by resolving a system of nonlinear equations. It is proved that if the underlying graph is strongly connected and the intersection of constraint intervals is empty, then there exists a unique equilibrium point; and if the intersection is nonempty, then there exist multiple equilibrium points, all of which lead to state consensus. We also establish several conditions for the local stability of the unique equilibrium point (corresponding to the case with empty intersection of constraint intervals) or local constraint consensus (corresponding to the case with nonempty intersection of constraint intervals) by invoking Lyapunov's indirect method. Characterization of the global behavior of such a second-order multiagent system is technically rather challenging at this stage. As a first step toward this end, we show in two special cases that global convergence to the unique equilibrium point or state consensus can be guaranteed by employing Lyapunov stability theory and robust analysis techniques. Finally, some numerical examples are provided to illustrate the theoretical findings.**

***Index Terms*—Directed network, existence and uniqueness of an equilibrium point, interval consensus, second-order multiagent systems (MASs).**

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I. Introduction

COORDINATION, in particular consensus seeking that aims to reduce the "distance" between different states, of multiagent systems (MASs) via local information exchange has been extensively addressed over the past decades [1]–[5], [8], [9], [11]. Various interesting issues have received a great amount of attention from researchers, for instance, the control protocol design and convergence analysis, and the applications of coordination algorithms to such relevant fields as signal processing and distributed optimization [6], [7].

In particular, the application of multiagent control to social groups for opinion formation is of special interest recently [3], [10]. Therefore, it is important to involve various features of social networks into the studies of multiagent control. A notable feature of social network is that constraint is ubiquitous. For example, agents may only interact with those close in opinion (as described in bounded confidence model) [12]–[14] and the interactions between the agents might be diffusive (such that the opinion of each agent updates as a function, e.g., an average, of its neighbors') [6], [15] or temporal [16]. In addition, in social networks, stubbornness and bias toward initial opinion [20]–[23], discrepancies in expressed and private opinions [24], and transmitting capacity of opinions [25] significantly influence how agents interact with each other and evolve. Apart from constraints on interactions, it is observed in consensus seeking that agents may tend to veto a certain consensus value or expect the consensus value to be within an admissible region [17], [18]. For instance, when the board members of a company are negotiating a sell order, each of them may impose boundaries on the price. The aim of the negotiation is to reach an agreement on price. In particular, a consensus value that respects everyone's interval constraints on price has priority over, for example, a simple average of the initial bid [18]. We are interested in this kind of interval constraint in the present work and will take it into consideration in agreement seeking of MASs.

Consensus control with state constraint has been investigated for MASs over the past few years, mainly for single-integrator dynamics. When imposing state constraint, the state of each agent is usually required to stay within an interval *all the time* [26], [27]. For instance, for the extremum seeking problem

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in smart grid, the generated powers are subjected to interval constraints since they cannot exceed the capability of generators [19]; in sensor networks, both flows over the channels and the amount of resource that each agent can store are subjected to capability limitation [28]. A simple discarded consensus algorithm is proposed in [26], where each agent discards the states of its neighbors that fall outside its own constraint interval. This algorithm keeps the state of each agent restricted in its own constraint interval and ensures consensus when these intervals admit nonempty intersection. From the perspective of potential function, Lee and Mesbahi [27] find that the gradient of such a function involving constraint terms yields a consensus protocol, which guarantees state constraint with appropriately chosen parameters. In [29], the authors investigate the consensus problem for nonlinear MASs with time-varying state constraints and full-rank coupling gain, which is transformed into a bounded consensus problem that is easier to deal with. Inspired by typical distributed convex optimization algorithms, projection-based method is also used in the state constrained consensus problem [7], [30]. In [7], the authors present a distributed projected consensus algorithm in which agents combine local averaging with projection on their individual constraint sets. Along this line, there have been numerous works on consensus optimization, see, e.g., [30].

Different from the works on *hard* constraints, which cannot be violated at any time, on system state reviewed earlier, a novel constrained consensus model for single-integrator MAS is recently proposed in [17] and [18]. The main idea is to introduce a set of constraint intervals for agent dynamics in which a consensus value can be accepted, and then force the consensus value to belong to the intersection of all such intervals. The constraint interval can represent each agent's preference or its heuristic attitude toward an issue, as indicated in the negotiation example of board members on price of a sell order. Moreover, the constraints are not hard on the states but *trespassable* during the transient evolution. This constrained consensus model adopts the diffusivelike control protocol except that the values exchanged are limited within a certain interval, i.e., saturated using a nonlinear function. In such a setup, if consensus of the MAS is realized, then it is called an *interval consensus* [17], [18].

In engineering applications and social networks, the second-order dynamics is usually introduced to model the evolution of agents, see, for instance, [31]. In this article, we aim to extend the work of interval consensus by Fortan *et al.* [17], [18] to the second-order MAS over a directed network. It is worthwhile to point out that although the majority of social opinion evolution studies focus on first-order dynamics, second-order (i.e., flocking) dynamics has its own significance, as is widely used in the classical Cucker–Smale model and Vicsek model [38]. Moreover, a second-order dynamics can also help to accelerate the convergence to an equilibrium point [39]. In particular, we consider the following two interval consensus frameworks.

1) *Agent Interval Consensus:* The node states are constrained in their own constraint intervals.
2) *Neighbor Interval Consensus:* The node states are constrained in their neighbors' constraint intervals.

These two frameworks can be viewed as two different metrics for social norms, where the opinions outside the interval are extreme ones. In particular, the agent interval consensus can be seen as that all agents hold persuasion bias [32] and fail to account for possible repetition in the information they receive. Hence, each agent repeatedly sends an opinion falling into its own constraint interval via saturation to persuade its neighbors to take an opinion belonging to this interval. The neighbor interval consensus can be interpreted as that all agents hold confirmation bias [33]. Specifically, each agent discounts information that undermines its past judgment, which is represented by the constraint interval, by projecting the neighbors' opinions into its own constraint interval to steer its opinion. The former framework includes that studied in [17] and [18] as a special case. The latter one, to the best of our knowledge, has not been studied in the literature. By treating the second framework as a "dual" framework of the first one, we can also gain insight into how constraint types affect the collective behavior of agents.

The purpose of this article is two-fold. First, we aim to specify all equilibrium points, where only nonempty intersection case is fully discussed by Fortan *et al.* [17], [18]. The challenge lies in resolving a system of nonlinear equations having saturation terms. Second, we aim to investigate the local and global stability of equilibrium point. Since the velocity term is involved in the second-order dynamics, the analysis developed in [17] and [18] by exploiting a certain "monotonic" property via an appropriately designed Lyapunov function is not applicable. Moreover, the existence of these two constraints makes the system dynamics being piecewise linear, and therefore most of the existing analysis techniques proposed for second-order dynamics without constraints do not work.

For the two different types of interval constraints in the agent dynamics, we work out the following results, which are also the main contributions of this article.

i) Equilibria Existence and Uniqueness: We prove that the equilibrium point exists uniquely if the network is strongly connected and the intersection of all constraint intervals is empty in both frameworks. To this end, we develop a novel analysis technique via induction and contradiction arguments to characterize the relation between agents' states and their saturated counterparts. This method differs from the one developed in [18], which uses the fixed point theorem. Moreover, the authors in [18] only prove the existence of an equilibrium point, whose uniqueness is not addressed and left as an open problem but tackled in this article.
ii) Local Stability: In the empty intersection case, we prove that if the unique equilibrium point satisfies that at this equilibrium point, each agent stays away from the boundary of certain intervals, then its local asymptotic stability can be guaranteed provided that the damping gain is larger than a threshold. This extends the local stability result of Fontan *et al.* [18] to the second-order case by carefully characterizing the spectrum of system matrix near the equilibrium point. In the nonempty intersection case, due to the lack of "monotonic" property in second-order dynamics, the proof for global convergence to

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consensus as in [18] is quite a challenge. We introduce a new concept of local constraint consensus (see Definition 3) to facilitate the stability analysis, based on which Lyapunov stability theory becomes applicable.

iii) Global Stability: By robust analysis technique and the translation invariance property (that is, a translation motion of x-axis changes agents' positions and constraint intervals but does not influence the convergence property), we prove that if all constraint intervals are the same and the network is strongly connected, then consensus can be realized globally asymptotically with a sufficiently large damping gain for both frameworks. Moreover, for neighbor interval consensus, if the constraint intervals are pairwise disjoint and the damping gain is larger than a threshold, then the unique equilibrium point is globally asymptotically stable.

In addition, we also show the difference between the two types of interval consensus. In particular, for the agent interval consensus, even the constraint intervals admit empty intersection, the equilibrium point can still be a consensus point, which, however, is impossible for the neighbor interval consensus. Also, due to heterogeneity of the constraint types, the local and global stability results are obtained for different kinds of equilibrium points or different distributions of constraint intervals in the two frameworks.

The rest of this article is organized as follows. The problem formulation is presented in Section II. The main results for interval consensus of two frameworks are presented in Section III and Section IV, respectively. Numerical examples are given in Section V to demonstrate our theoretical results. Finally, Section VI concludes this article. The proofs are put in the Appendix.

Notation: Let $\mathrm{diag}\{a_1,\ldots,a_N\}$ represent a diagonal matrix whose ith diagonal entry is a_i for $i=1,\ldots,N$. We use $\mathsf{Re}(c)$ and $\mathsf{Im}(c)$ to denote, respectively, the real and imaginary parts of a complex number c. $\lambda_i(\mathbf{A})$ denotes the ith eigenvalue of $\mathbf{A}\in\mathbb{R}^{N\times N}$ for $i=1,\ldots,N$ (the eigenvalues of $\mathbf{A}$ are arranged in a certain order). We use $\mathrm{con}\{\mathscr{A}\}$ to represent the convex hull of the set $\mathscr{A}$, i.e., $\mathrm{con}\{\mathscr{A}\}=\{\sum_{j=1}^{p}\alpha_j x_j|\sum_{j=1}^{p}\alpha_j=1,\alpha_j\geq 0, x_j\in\mathscr{A}$ for all j, and p is any positive integer$\}$. $\mathrm{Int}(\mathscr{A})$ is the set of interior points of $\mathscr{A}$. Let $\|\cdot\|$ denote the Euclidean norm of a finite dimensional vector or spectral norm of a real matrix of finite order. Denote by $\mathbf{I}_n$ the identity matrix and by $\mathbf{0}_{n\times n}$ the zero matrix in $\mathbb{R}^{n\times n}$ (if the subscript is dropped, $\mathbf{I}$ and $\mathbf{0}$ have compatible dimensions). We use $\mathbf{M}>\mathbf{0}$ to denote that a square matrix $\mathbf{M}$ is positive definite.

Graph Theory Basics (see [34] and [35]): Let $\mathcal{G}=(\mathsf{V},\mathsf{E},\mathsf{A})$ be a weighted directed graph of order N with a finite nonempty set of nodes $\mathsf{V}=\{1,2,\ldots,N\}$ and a set of edges $\mathsf{E}\subset\mathsf{V}\times\mathsf{V}$. If $(i,j)\in\mathsf{E}$, then j can receive the information from i. A weighted adjacency matrix $\mathsf{A}=[a_{ij}]\in\mathbb{R}^{N\times N}$ is defined such that a_{ij} is the weight of the directed edge (j,i) and $a_{ij}>0$ if $(j,i)\in\mathsf{E}$; $a_{ij}=0$ otherwise. Assume that no self-loop exists, i.e., $a_{ii}=0$ for all $i\in\mathsf{V}$. If $(j,i)\in\mathsf{E}$, j is said to be a neighbor of i. Denote by $\mathcal{N}_i=\{k|(k,i)\in\mathsf{E}\}$ the set of node i's neighbors in $\mathcal{G}$. The Laplacian matrix $\mathbf{L}$ of $\mathcal{G}=(\mathsf{V},\mathsf{E},\mathsf{A})$ is defined as $\mathbb{R}^{N\times N}\ni\mathbf{L}=[\ell_{ij}]=\mathrm{diag}\{d_1,\ldots,d_N\}-\mathsf{A}$, where $d_i=\sum_{j=1}^{N}a_{ij}$ is the degree of node i, $i=1,\ldots,N$. A directed path from node i to node j in the graph $\mathcal{G}$ is a sequence of edges of the following form: $(i,p_1),(p_1,p_2),\ldots,(p_n,j)\in\mathsf{E}$. If between any distinct nodes i and j of $\mathcal{G}$, there is a directed path from i to j, then we call $\mathcal{G}$ strongly connected.

II. Problem Definition

In this section, we introduce the system model and define the problems of interest.

A. Interval Consensus for Second-Order MAS

Consider a networked MAS with N agents whose evolution is governed by

$$\begin{cases}\dot{\mathbf{x}}_i(t)=\mathbf{v}_i(t)\\ \dot{\mathbf{v}}_i(t)=-k\mathbf{v}_i(t)+\sum\limits_{j\in\mathcal{N}_i}a_{ij}\left(\mathbf{H}_{ij}\left(\mathbf{x}_j(t)\right)-\mathbf{x}_i(t)\right)\end{cases}\tag{1}$$

for $i=1,\ldots,N$, where by following the convention, $\mathbf{x}_i\in\mathbb{R}$ is termed position and $\mathbf{v}_i\in\mathbb{R}$ the velocity of the ith agent, $k>0$ is a damping gain, and $\mathbf{H}_{ij}(\cdot):\mathbb{R}\to\mathbb{R}$ is a continuous nonlinear function.

Let χ_m be defined as

$$\chi_m(z)=\begin{cases}q_m,\ \text{if } z>q_m\\ z,\ \text{if } p_m\leq z\leq q_m\\ p_m,\ \text{if } z<p_m\end{cases}$$

where p_m and q_m are real numbers and $p_m<q_m$, $m=1,\ldots,N$. Denote by $\mathcal{I}_m$ the closed interval $[p_m,q_m]$, i.e., $\mathcal{I}_m=[p_m,q_m]$, $m=1,\ldots,N$.

We consider in this article the following two frameworks.

1) *Agent Interval Consensus [18]:* There holds $\mathbf{H}_{ij}(\cdot)=\chi_j(\cdot)$ for all $i,j\in\mathsf{V}$, i.e., the node states are constrained in their own constraint intervals.
2) *Neighbor Interval Consensus:* There holds $\mathbf{H}_{ij}(\cdot)=\chi_i(\cdot)$ for all $i,j\in\mathsf{V}$, i.e., the node states are constrained in their neighbors' constraint intervals.

Remark 1: A simplified accelerated differential equation (ADE) [39] can be written as: $\ddot{\mathbf{x}}+\gamma(t)\dot{\mathbf{x}}+\nabla\mathbf{R}(\mathbf{x})=0$ with appropriate $\gamma(t)>0$. The ADE is actually a variant of Nesterov accelerate gradient flow and accelerates the convergence of $\mathbf{x}(t)$ to an optimal point [39]. Under certain conditions, an optimal point in ADE might be the solution to $\nabla\mathbf{R}(\mathbf{x})=\mathbf{0}$. If the second-order dynamics (1) is written into a second-order differential equation, then $\nabla\mathbf{R}(\mathbf{x})=\mathbf{0}$ is $\mathbf{F}(\mathbf{x})=\mathbf{0}$ with the ith component of $\mathbf{F}(\mathbf{x})$ being $-\sum_j a_{ij}(\mathbf{H}_{ij}(\mathbf{x}_j(t))-\mathbf{x}_i(t))$. The solution of $\mathbf{F}(\mathbf{x})=\mathbf{0}$ gives the equilibrium point (see Section II-D). Hence, the use of a second-order dynamics may help accelerate the convergence of the solution of (1) to the equilibrium point.

B. Motivating Examples

The following examples are first proposed in [17] and [18]. They motivate the investigation of constrained consensus such

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that a consensus value has to respect certain constraints. However, there is no need for the transient dynamics to respect them, i.e., the constraints are soft.

1) *Load balance with resource constraints:* In some cases, a group of computational units must share in equal parts a certain workload, under the constraint that each unit can provide only a certain amount of resources, not known *a priori* to the other units. A feasible workload balance must respect the resource constraints.
2) *Achieving a price agreement:* The board members of a company are negotiating a buy or sell order, and seek an agreement among themselves on a price. If unanimity of the board is required, then a consensus value that respects everybody's constraints has priority over for instance a simple average of the initial bids.

C. Model Interpretations

The term $\mathbf{H}_{ij}(\cdot)$ means that the information exchange is constrained by this nonlinear function. Such a constraint is useful in social networks to capture individual biases [17], [18]. Note that an interval constraint can be viewed as a metric for social norms, where opinions outside the interval are extreme ones. The social interpretation of the two ways of placing soft constraints in (1), i.e., agent interval consensus and neighbor interval consensus, can be described as follows.

1) *Agent Interval Consensus:* The condition that $\mathbf{H}_{ij}(\cdot) = \chi_j(\cdot)$ for all i and j implies that every node k in the network intends to hide $\mathbf{x}_k$ and reveals only $\chi_k(\mathbf{x}_k)$ to its neighbors. The interval $[p_k, q_k]$ is therefore a type of social norm from the self-perspective of agent k, where the true $\mathbf{x}_k$ may be too socially polarized to be expressed. In social terms, this may be explained as the influence of peer and group pressure [24], or observer's effect in publicly disclosed opinions [17], [18].
2) *Neighbor Interval Consensus:* The condition that $\mathbf{H}_{ij}(\cdot) = \chi_i(\cdot)$ for all i and j can be interpreted as that nodes always reveal their true opinions to their neighbors, but each node holds personal bias against others' views. For agent i, the interval $[p_i, q_i]$ reflects a social norm that shows how agent i is willing to accept others' opinions. If the opinions of agent i's neighbors lie outside its constraint interval, then they are projected into this interval. Therefore, the interval $[p_i, q_i]$ is a type of confirmation bias where each agent discounts information that undermines its past judgments [32], [33].

D. Problems of Interest

Let $\mathbf{z} = [\mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{v}_1, \ldots, \mathbf{v}_N]^\top$ be the stacked state vector of (1). Assume that $\mathbf{z}^* = [\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*, \mathbf{v}_1^*, \ldots, \mathbf{v}_N^*]^\top$ is an equilibrium point. Obviously, $\mathbf{v}_i^* = 0$ for all i. Hence, to specify all the equilibrium points of (1), it suffices to solve the following system of nonlinear equations:

$$\sum_{j\in\mathcal{N}_i} a_{ij}\left(\mathbf{H}_{ij}(\mathbf{x}_j) - \mathbf{x}_i\right) = 0, i = 1, \ldots, N. \tag{2}$$

The goal of this article is two-fold. First, we aim to prove the existence and uniqueness of the equilibrium point, viz., to characterize the solution of nonlinear equations (2). Second, we aim to investigate the local and global stability of the equilibrium point. For subsequent analysis, we provide the definitions of an interior equilibrium point (IEP) and an equi-IEP (E-IEP), and the stability of an equilibrium point.

Definition 1: An equilibrium point $\mathbf{z}^* = [\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*, \mathbf{0}_{1\times N}]^\top$ is said to be an IEP of system (1) if for each i, $\mathbf{x}_i^*$ is either an interior point of $[p_i, q_i]$ or an interior point of $\mathbb{R} \setminus [p_i, q_i]$. In addition, $\mathbf{z}^*$ is said to be an E-IEP of system (1) if for each $j \in \mathsf{V}$, for any $i \in \mathcal{N}_j$, $\mathbf{x}_i^*$ is either an interior point of $[p_j, q_j]$ or an interior point of $\mathbb{R} \setminus [p_j, q_j]$.

Remark 2: It is worth pointing out that IEP and E-IEP are not mutually exclusive. That is to say, an equilibrium point can be an IEP and also an E-IEP.

Definition 2 (cf. [36]): Consider the autonomous system $\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y})$, where $\mathbf{f}$ is a Lipschitz map and $\mathbf{f}(\mathbf{0}) = \mathbf{0}$. The equilibrium point $\mathbf{y} = \mathbf{0}$ is Lyapunov stable if, for each $\epsilon > 0$, there is $\delta = \delta(\epsilon) > 0$ such that $\|\mathbf{y}(0)\| < \delta \Rightarrow \|\mathbf{y}(t)\| < \epsilon$ for all $t \geq 0$; asymptotically stable if it is stable and δ can be chosen such that $\|\mathbf{y}(0)\| < \delta \Rightarrow \lim_{t\to\infty} \mathbf{y}(t) = \mathbf{0}$; if in addition $\delta = \infty$, then $\mathbf{y} = \mathbf{0}$ is said to be globally asymptotically stable.

Remark 3: Note that the saturation function $\chi_j(\cdot)$ in (1) is not differentiable at p_j or q_j for all $j \in \mathsf{V}$. Therefore, the concepts of IEP and E-IEP facilitate the application of Lyapunov's indirect method to stability analysis.

III. Agent Interval Consensus

We start by investigating the existence and uniqueness of the equilibrium point for system (1). The following theorem says that if $\cap_{j\in\mathsf{V}}\mathcal{I}_j = \emptyset$, then there exists a unique equilibrium point, which confirms one conjecture in [18].

Theorem 1: Consider the agent interval consensus with $\mathbf{H}_{ij} = \chi_j$, $i, j = 1, \ldots, N$. Suppose that $\mathcal{G}$ is a directed and strongly connected graph. Then, there holds for system (1) the following results.

i) If $\cap_{j=1}^N \mathcal{I}_j = \emptyset$, then there exists a unique equilibrium point.
ii) If $\cap_{j=1}^N \mathcal{I}_j \neq \emptyset$, then there are multiple equilibrium points and each equilibrium point takes the form of $[\mathbf{1}_N^\top \cdot c, \mathbf{0}_{1\times N}]^\top$ with $c \in \cap_{j=1}^N \mathcal{I}_j$.

Remark 4: Theorem 1 extends the result of Fontan *et al.* [18] from first-order to second-order dynamics. Note that the uniqueness of the equilibrium point for the empty intersection case, as is stated in Theorem 1(i), is not addressed and left open in [18] but is successfully proved in our work by proposing a novel analysis technique. With this novel analysis technique, we can also analyze the equilibrium points for the case with nonempty intersection, as is stated in Theorem 1(ii). Theorem 1 provides a full picture regarding the existence and uniqueness of equilibrium point.

Remark 5: It is worthwhile pointing out that for the empty intersection case, i.e., $\cap_{j=1}^N \mathcal{I}_j = \emptyset$, the unique equilibrium point $\mathbf{z}^*$ may still satisfy that $\mathbf{x}_i^* = \mathbf{x}_j^*$ for all i, j, which implies that all agents reach consensus. This is the consequence of the social

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norm described in the model interpretations, where although $\mathbf{x}_j$ is hidden and $\chi_j(\mathbf{x}_j)$ is exchanged over the network, each agent is influenced by its neighbors and is able to leave its own preferred interval. A further discussion is referred to Example 3 in Section V.

A complete proof of Theorem 1 is referred to Appendix A. Moreover, in what follows, for presentation's sake, we put all the proofs in the Appendix.

According to Theorem 1, for the nonempty intersection case that $\cap_j \mathcal{I}_j \neq \emptyset$, there exist multiple equilibrium points and they are all consensus points. To study the stability of these equilibrium points in the presence of constraints, we need the following definition of local constraint consensus. Let $\mathbf{w} = [\mathbf{w}_1, \ldots, \mathbf{w}_N]^\top$ be an left eigenvector of $\mathbf{L}$ associated with eigenvalue zero satisfying that $\mathbf{w}^\top \mathbf{1}_N = 1$. A simple fact of $\mathbf{w}$ is that $\mathbf{w}_i > 0, i = 1, \ldots, N$ if $\mathcal{G}$ is strongly connected [34, Lemma 13.9.1], [35, Lemma 2.28].

Definition 3 (Local Constraint Consensus): Consensus is said to be realized locally for system (1) with constraint $\mathbf{H}_{ij}$, $i, j = 1 \ldots, N$, if for any closed set $\mathscr{B} \subset \mathrm{Int}(\cap_{j=1}^N \mathcal{I}_j)$, there exists a $\delta > 0$ such that whenever $\mathbf{x}_i(0) \in \mathscr{B}$, $\mathbf{v}_i(0) = 0$ for all i, and $\|\mathbf{z}(0) - \mathbf{z}^*\| \leq \delta$, one has $\mathbf{z}(t) \to \mathbf{z}^*$ as $t \to \infty$, where $\mathbf{z}^* = [\mathbf{w}^\top \mathbf{x}(0) \mathbf{1}_N^\top, \mathbf{0}_{1\times N}]^\top$.

Remark 6: Local constraint consensus requires that $\mathbf{x}_i(0) \in \mathscr{B}$ and $\mathbf{v}_i(0) = 0$ for all $i \in \mathsf{V}$. If this condition is removed, local constraint consensus turns to be local consensus. This concept enables us to perform consensus analysis by ensuring that all agents will not move across the boundaries of constraint intervals.

Due to existence of $\chi_j(\cdot)$ in (1), the following matrix is crucial for analysis of local stability of a unique IEP $\mathbf{z}^*$:

$$\hat{\mathbf{L}}_1 = \mathbf{L}_1 + \mathrm{diag}\left\{\sum_{j\in\Lambda} a_{1j}, \ldots, \sum_{j\in\Lambda} a_{Nj}\right\}$$

where $\mathbf{L}_1$ is a Laplacian matrix of $\mathcal{G}_1$, a subgraph of $\mathcal{G}$ obtained by removing any edge $(s, \ell) \in \mathsf{E}$ satisfying $s \in \Lambda$ with $\Lambda = \{j \,|\, \mathbf{x}_j^* \notin \mathcal{I}_j\}$. With Definitions 1–3 and $\hat{\mathbf{L}}_1$ in place, we are ready to present the results for local stability of a unique equilibrium point via characterizing the damping gain k and local constraint consensus.

Theorem 2: Consider the agent interval consensus with $\mathbf{H}_{ij} = \chi_j$, $i, j = 1, \ldots, N$. Assume that $\mathcal{G}$ is directed and strongly connected. Suppose that $\mathbf{z}^* = [\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*, \mathbf{0}_{1\times N}]^\top$ is an IEP of system (1), then there holds the following results.

i) If $\cap_j \mathcal{I}_j = \emptyset$ and $\lambda_i(\hat{\mathbf{L}}_1)$ has nonzero imaginary part for some $i \in \mathsf{V}$, then there exists $k^* > 0$ such that any IEP is locally asymptotically stable if the damping gain $k > k^*$, whereas it is unstable if $0 < k < k^*$.

ii) If $\cap_j \mathcal{I}_j = \emptyset$ and $\lambda_i(\hat{\mathbf{L}}_1)$ has zero imaginary part for any $i \in \mathsf{V}$, then any IEP is locally asymptotically stable for any damping gain $k > 0$.

iii) If $\mathrm{Int}(\cap_j \mathcal{I}_j) \neq \emptyset$, then there exists $\hat{k} \geq 0$ such that local constraint consensus can be realized for any damping gain $k > \hat{k}$.

Remark 7: It is worth pointing out that an equilibrium point $\mathbf{z}^*$ is likely to not be an IEP. In this case, there exists an $i \in \mathsf{V}$ such that $\mathbf{x}_i^* \in \{p_i, q_i\}$. Hence, if $\mathbf{z}(t)$ lies in a small vicinity of $\mathbf{z}^*$, then system dynamics might change due to the existence of χ_i. As a result, the stability analysis of a non-IEP is much more difficult than an IEP.

Remark 8: In Theorem 2, we use linearization technique for local stability analysis. The location of eigenvalue λ_i of $\hat{\mathbf{L}}_1$ determines the stability of the linearized system dynamics. The two cases (i) and (ii) concerning λ_i arise due to spectrum analysis.

For empty intersection, the local stability results concerning IEP in Theorem 2(i) and (ii) extend that for the first-order dynamics in [18] to the second-order case, by indirect method of Lyapunov to characterize the system matrix around the equilibrium point. For nonempty intersection case dealt with in Theorem 2(iii), since a velocity term is involved in the system dynamics, the analysis technique to exploit a certain "monotonic" property in [18] does not work for the second-order dynamics. To analyze the stability of equilibrium points in this case, we introduce the concept of local constraint consensus, which facilitates the stability analysis based on Lyapunov stability theory.

With the results in Theorem 2(i) and (ii) in mind, a natural question arises as to under what conditions an equilibrium point turns out to be an IEP. If $\cap_{j\in\mathsf{V}} \mathcal{I}_j = \emptyset$, the following proposition gives a mild condition to verify whether the unique equilibrium point is also an interior one.

Proposition 1: Consider the agent interval consensus with $\mathbf{H}_{ij} = \chi_j$, $i, j = 1, \ldots, N$ and $\mathcal{G}$ being strongly connected. If $\tilde{\mathcal{I}}_m \cap \mathcal{I}_m = \emptyset$ or $\tilde{\mathcal{I}}_m \subset \mathrm{Int}(\mathcal{I}_m)$ for all $m \in \mathsf{V}$, then each equilibrium point of system (1) is also an IEP, where $\tilde{\mathcal{I}}_m = \left[\sum_{j\in\mathcal{N}_m} \alpha_m^j p_j, \sum_{j\in\mathcal{N}_m} \alpha_m^j q_j\right]$ and $\alpha_i^j = \frac{a_{ij}}{\sum_{m\in\mathcal{N}_i} a_{im}}$, $i, j = 1, \ldots, N$.

In the following, we show that if $\mathcal{G}$ is strongly connected and the constraint intervals are the same for all agents, then global asymptotic convergence to consensus can be guaranteed with an appropriate damping gain k.

Theorem 3: Consider the agent interval consensus with $\mathbf{H}_{ij} = \chi_j$, $i, j = 1, \ldots, N$. Assume that $\mathcal{G}$ is strongly connected, and further $\mathcal{I}_m = \mathcal{I}_n$ for all $m, n \in \mathsf{V}$. Then, consensus can be realized globally and asymptotically for system (1) if the damping gain k is sufficiently large.

It is worth pointing out that in [18], it is proved that consensus can be achieved globally asymptotically without the requirement that all constraint intervals are the same. The obstacle to prove global stability in a more general case $\cap_{j=1}^N \mathcal{I}_j \neq \emptyset$ in the present work lies in (i) the lack of "monotonicity" as exploited in [18] for single-integrator dynamics, and (ii) the piecewise linear form of (1) caused by saturation functions.

IV. Neighbor Interval Consensus

In this section, we consider the framework $\mathbf{H}_{ij} = \chi_i$, $i, j = 1, \ldots, N$. The following theorem presents the results concerning existence and uniqueness of the equilibrium point with a strongly connected graph, which resemble or parallel those in Theorem 1. Different from Theorem 1, Theorem 4 assures that: 1) any equilibrium point $\mathbf{z}^*$ satisfies that $\mathbf{x}_i^* \in \mathcal{I}_i$ for all $i \in \mathsf{V}$; and 2)

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in the pairwise disjoint case, i.e., $\mathcal{I}_p \cap \mathcal{I}_q = \emptyset$ for all p, q, the equilibrium point can be calculated explicitly.

Theorem 4: Consider the neighbor interval consensus with $\mathbf{H}_{ij} = \chi_i$, $i, j = 1, \ldots, N$. Assume that $\mathcal{G}$ is strongly connected. Then, there holds for system (1) the following results.

i) If $\cap_{j=1}^N \mathcal{I}_j = \emptyset$, then there exists a unique equilibrium point $\mathbf{z}^* = [(\mathbf{x}^*)^\top, \mathbf{0}_{1\times N}]^\top$ and $\mathbf{x}_i^* \in \mathcal{I}_i$ for all $i \in \mathsf{V}$.

ii) If $\cap_{j=1}^N \mathcal{I}_j \neq \emptyset$, then there exist multiple equilibrium points and each takes the form of $[\mathbf{1}_N^\top \cdot c, \mathbf{0}_{1\times N}]^\top$ with $c \in \cap_{j=1}^N \mathcal{I}_j$.

iii) If $\mathcal{I}_m \cap \mathcal{I}_n = \emptyset$ for any pair $m, n \in \mathsf{V}$, then the unique equilibrium point $\mathbf{z}^* = [(\mathbf{x}^*)^\top, \mathbf{0}_{1\times N}]^\top$ is such that $\mathbf{x}^* = \mathbf{A}^{-1}\mathbf{b}$ with $\mathbf{A} = \text{diag}\{\sum_{j\in\mathcal{N}_1} a_{1j}, \ldots, \sum_{j\in\mathcal{N}_N} a_{Nj}\} > \mathbf{0}$ being the degree matrix, and $\mathbf{b} = \text{diag}\{\sum_{j\in\mathcal{N}_1} a_{1j} y_j^1, \ldots, \sum_{j\in\mathcal{N}_N} a_{Nj} y_j^N\}$, where for any $i \in \mathsf{V}$, given $j \in \mathcal{N}_i$, $y_j^i = p_i$ if $q_j \leq p_i$ or $y_j^i = q_i$ if $p_j \geq q_i$.

It is easily observed from Theorem 4 that the equilibrium point is also a consensus point only if $\cap_{j=1}^N \mathcal{I}_j \neq \emptyset$. This is because $\mathbf{x}_i^*$ in the unique equilibrium point $\mathbf{z}^*$ satisfies that $\mathbf{x}_i^* \in \mathcal{I}_i$ for all i if $\cap_{j=1}^N \mathcal{I}_j = \emptyset$. The aforementioned observation can be interpreted from the perspective of social norm. Each agent is biased against its neighbors' opinions in neighbor-interval-consensus framework and would like to stay in its own preferred interval. This differs from the result in Theorem 1, where even if $\cap_{j=1}^N \mathcal{I}_j = \emptyset$, the unique equilibrium point can still be a consensus point.

Similar to Theorem 2, we need the following matrix for stability analysis when a unique equilibrium point $\mathbf{z}^*$ exists:

$$\hat{\mathbf{L}}_2 = \mathbf{L}_2 + \text{diag}\left\{\sum_{j\in\Lambda_1} a_{1j}, \ldots, \sum_{j\in\Lambda_N} a_{Nj}\right\}$$

where $\mathbf{L}_2$ is a Laplacian matrix of $\mathcal{G}_2$, a subgraph of $\mathcal{G}$ obtained by removing any edge $(s, \ell) \in \mathsf{E}$ satisfying $s \in \Lambda_\ell$ with $\Lambda_\ell = \{j \,|\, \mathbf{x}_j^* \notin \mathcal{I}_\ell\}$ for $\ell = 1, \ldots, N$. We now present the theorem concerning local stability of a unique equilibrium point via characterizing the damping gain k or local constraint consensus.

Theorem 5: Consider the neighbor interval consensus with $\mathbf{H}_{ij} = \chi_i$, $i, j = 1, \ldots, N$. Assume that $\mathcal{G}$ is strongly connected. Let $\mathbf{z}^* = [\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*, \mathbf{0}_{1\times N}]^\top$ be an equilibrium point. Then, the following statements hold for the system (1).

i) Suppose that $\cap_{j=1}^N \mathcal{I}_j = \emptyset$, $\mathbf{z}^*$ is an E-IEP, and $\lambda_i(\hat{\mathbf{L}}_2)$ has nonzero imaginary part for some $i \in \mathsf{V}$. There exists a $k^* > 0$ such that if the damping gain $k > k^*$, then $\mathbf{z}^*$ is locally asymptotically stable, whereas if $k < k^*$, then $\mathbf{z}^*$ is unstable.

ii) Suppose that $\cap_{j=1}^N \mathcal{I}_j = \emptyset$, $\mathbf{z}^*$ is an E-IEP, and $\lambda_i(\hat{\mathbf{L}}_2)$ has zero imaginary part for all $i \in \mathsf{V}$. Then, for any damping gain $k > 0$, $\mathbf{z}^*$ is locally asymptotically stable.

iii) If $\text{Int}(\cap_{j=1}^N \mathcal{I}_j) \neq \emptyset$, then there exists $\hat{k} \geq 0$ such that local constraint consensus can be realized for any damping gain $k > \hat{k}$ with $\mathbf{x}_j^* \in \cap_{m=1}^N \mathcal{I}_m$ for all $j \in \mathsf{V}$.

The following proposition presents a simple condition to verify that whether an equilibrium point is an E-IEP.

Proposition 2: Consider the neighbor interval consensus with $\mathbf{H}_{ij} = \chi_i$ for all $i, j = 1, \ldots, N$. Assume that $\mathcal{G}$ is strongly connected. Suppose that for any i, either $\mathcal{I}_j \cap \mathcal{I}_i = \emptyset$ or $\mathcal{I}_i \subset \text{Int}(\mathcal{I}_j)$ holds for all $j \in \mathcal{N}_i$. Then, any equilibrium point is an E-IEP of the system (1).

Finally, we show that global stability of the unique equilibrium point can be guaranteed if the constraint intervals are pairwise disjoint (which turns out to yield an E-IEP according to Proposition 2).

Theorem 6: Consider the neighbor interval consensus with $\mathbf{H}_{ij} = \chi_i$, $i, j = 1, \ldots, N$. Assume that $\mathcal{G}$ is strongly connected. Suppose that given any $j \neq k \in \mathsf{V}$, we have $\mathcal{I}_j \cap \mathcal{I}_k = \emptyset$. Let $\mathbf{z}^* = [\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*, \mathbf{0}_{1\times N}]^\top$ be an equilibrium point. Then, $\mathbf{z}^*$ is globally asymptotically stable for the system (1) if the damping gain k is sufficiently large.

It is worth pointing out that if $\mathcal{I}_j = \mathcal{I}_i$, then $\chi_j = \chi_i$ for any $j, i \in \mathsf{V}$. Therefore, the result established in Theorem 3 also applies to the case studied in Theorem 6. However, if given $j \in \mathsf{V}$, $\mathcal{I}_j \cap \mathcal{I}_i = \emptyset$ for all $i \neq j$, then Theorem 3 may no longer hold.

V. Numerical Examples

In this section, we present three numerical examples to illustrate the theoretical findings. Examples 1 and 2 are provided to illustrate the two different interval consensus frameworks, respectively, whereas Example 3 shows the difference between the distributions of equilibrium points in the two frameworks.

Example 1: This example is designed to verify the theoretical results for agent interval consensus, i.e., $\mathbf{H}_{ij} = \chi_j, i, j \in \mathsf{V}$. Consider a network of six agents interacting over directed graph $\mathcal{G}$, which is strongly connected and whose adjacency matrix is

$$\mathsf{A} = \begin{bmatrix} 0 & 0 & 0 & 0.5469 & 0 & 0 \\ 0.8147 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.7469 & 0 & 0 & 0 & 0.1270 \\ 0 & 0 & 0.9143 & 0 & 0.0975 & 0 \\ 0 & 0.5469 & 0.2843 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.6324 & 0.0524 & 0 \end{bmatrix}$$

The constraint intervals $\mathcal{I}_j, j = 1, \ldots, N$, are shown in Fig. 1 In the empty intersection case, the unique equilibrium point is $\mathbf{z}^* = [-4, 6, -1.6637, -1.6407, -1.4244, -3.8029, \mathbf{0}_{1\times 6}]^\top$, which is an IEP. Hence, $\Lambda = \{1, 2, 4\}$. It can be easily verified that $\hat{\mathbf{L}}_1$ has complex eigenvalues: $1.0118, 0.6418, 0.8741 \pm 0.0903j, 0.5469, 0.8147$. With $k = 0.9$, the IEP $\mathbf{z}^*$ is locally asymptotically stable, as is shown in Fig. 2 . If choosing a smaller k, say $k = 0.1$, it is shown in Fig. 3 that the IEP $\mathbf{z}^*$ is unstable. In the nonempty intersection case, local constraint consensus is seen to be achieved asymptotically with $k = 0.5$, which is depicted in Fig. 4 .

In this example, we also consider the case where all the constraint intervals are the same, i.e., $\mathcal{I}_m = [-1, 2]$ for all m. Choose $k = 5$. It can be observed from Fig. 5 that consensus can be realized from different initial states, indicating global convergence to state consensus.

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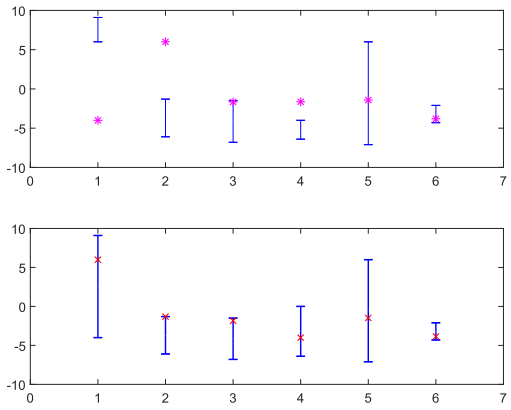


Fig. 1. x-axis is the index of the agent. The upper plot: constraint intervals $\mathcal{I}_1 = [6, 9.1]$, $\mathcal{I}_2 = [-6.1, -1.3]$, $\mathcal{I}_3 = [-6.8, -1.5]$, $\mathcal{I}_4 = [-6.4, -4]$, $\mathcal{I}_5 = [-7.1, 6]$, and $\mathcal{I}_6 = [-4.3, -2.1]$. It is clear in this case that $\cap_{j=1}^{N}\mathcal{I}_j = \emptyset$. The lower plot: constraint intervals $\mathcal{I}_1 = [-4, 9.1]$, $\mathcal{I}_2 = [-6.1, -1.3]$, $\mathcal{I}_3 = [-6.8, -1.5]$, $\mathcal{I}_4 = [-6.4, 0]$, $\mathcal{I}_5 = [-7.1, 6]$, and $\mathcal{I}_6 = [-4.3, -2.1]$, for which $\cap_{j=1}^{6}\mathcal{I}_j = [-4, -2.1] \neq \emptyset$. "$*$" and "$\times$" show the value of $\mathbf{x}_i^*$ in Example 1 and Example 2, respectively, for $i \in \mathsf{V}$.

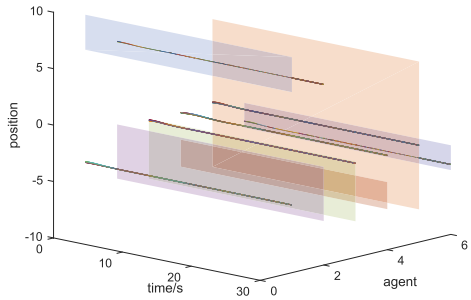


Fig. 2. Initial states are chosen from the small vicinity of $\mathbf{z}^*$. The faded regions represent the constraint intervals for all the agents, respectively. It is seen that $\lim_{t\to\infty}\mathbf{x}_i(t) = \mathbf{x}_i^*$ and $\lim_{t\to\infty}\mathbf{v}_i = 0$ with $k = 0.9$ for all $i = 1, \ldots, N$, showing local stability of $\mathbf{z}^*$.

Example 2: This example is designed to verify the theoretical results for neighbor interval consensus, i.e., $\mathbf{H}_{ij} = \chi_i$, $i, j \in \mathsf{V}$. We still use the directed graph and the constraint intervals in Example 1. In the empty intersection case, the unique equilibrium point is $\mathbf{z}^* = [6, -1.3, -1.8422, -4, -1.4856, -3.8546, \mathbf{0}_{1\times 6}]^\top$. Moreover, $\hat{\mathbf{L}}_2$ has real eigenvalues: 0.5469, 0.8312, 0.8147, 0.8739, 0.6848, 1.0118, and the equilibrium point is an E-IEP, which is locally stable with $k = 0.5$, as is shown in Fig. 6 . In the nonempty intersection case, local constraint consensus is observed to be reached asymptotically, as shown in Fig. 7 , with also $k = 0.5$.

To verify Theorem 6, set $\mathcal{I}_i = [2 \cdot (i-1), 2 \cdot i - 1]$ for $i = 1, \ldots, 6$. Evidently, $\mathcal{I}_j \cap \mathcal{I}_k = \emptyset$ for all $j \neq k$. Starting from different initial values, it is observed from Fig. 8 that the unique equilibrium point is globally asymptotically stable for $k = 5$.

Example 3: Inspired by Remark 5, we provide a brief discussion on whether we can configure the communication graph for reaching consensus with $\mathbf{H}_{ij} = \chi_j, i, j \in \mathsf{V}$, in the case $\cap_{j=1}^{N}\mathcal{I}_j = \emptyset$. Consider a complete graph with five nodes.

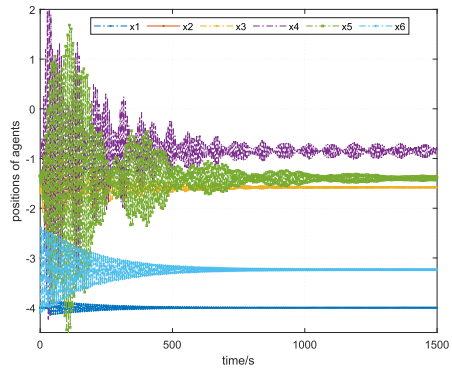


Fig. 3. Initial state is chosen from the small vicinity of $\mathbf{z}^*$. The positions of five agents do not converge with $k = 0.1$, showing instability of $\mathbf{z}^*$ if the damping gain is small when $\hat{\mathbf{L}}_1$ has complex eigenvalues.

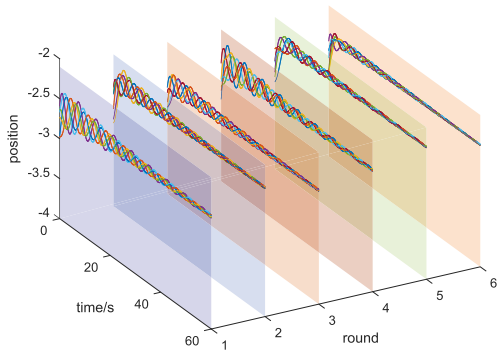


Fig. 4. Faded regions represent the nonempty intersection of constraint intervals. In different rounds, we choose different initial states. In each round, different colors indicate different state components. Choosing different initial points, the positions of six agents all stay in $[p_*, q_*]$, where $p_* = \max_j\{p_j\}$ and $q_* = \min_j\{q_j\}$, at any time and finally reach a consensus value.

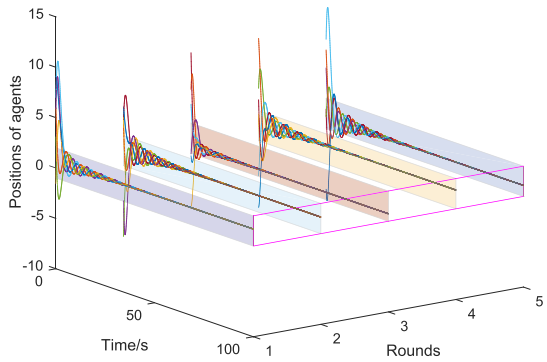


Fig. 5. Faded regions represent the same constraint interval for all the agents. In different rounds, we choose different initial states. In each round, different colors indicate different state components. With different initial states and $k = 5$, all agents reach a consensus value belonging to $[-1, 2]$.

The five intervals are set to be $\mathcal{I}_j = [2 \cdot (j-1), 2 \cdot (j-1) + 1]$, $j = 1, \ldots, 5$. It is clear that their intersection is empty. Next, we compute the edge weights to ensure that the unique equilibrium point is $[4.5 \times \mathbf{1}_N^\top, \mathbf{0}_{1\times 5}]^\top$. As a result, $\mathbf{x}_i^* = 4.5$ for all i. Take agent one for example. For $\mathbf{x}_1^* = 4.5$, one obtains

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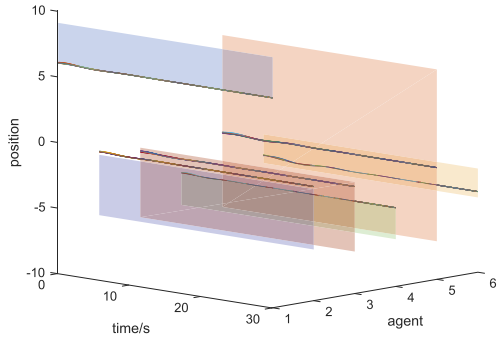


Fig. 6. Faded regions represent the constraint intervals for all the agents. Choose different initial states from the small vicinity of the equilibrium point $\mathbf{z}^*$. It is observed that all agents converge to this equilibrium point asymptotically.

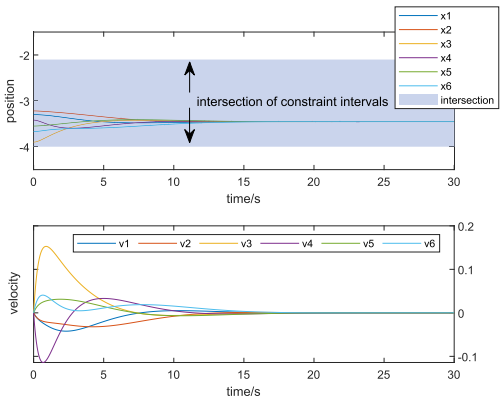


Fig. 7. Constraint intervals admit a nonempty intersection showed by the faded region. Starting from an interior point of a small vicinity of E-IEP, each agent, say i, stays in (p_i, q_i) and finally all agents reach the same position.

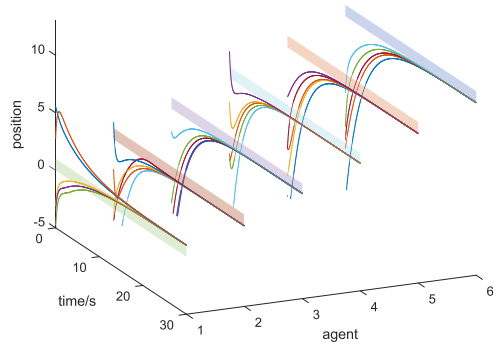


Fig. 8. Constraint interval for agent i is set to be $[2\cdot(i-1), 2\cdot i-1]$ for $i=1,\ldots,6$. Therefore, $\cap_{j=1}^{6}\mathcal{I}_j=\emptyset$. A unique equilibrium point exists. Starting from various different initial values, each agent arrives at the same position asymptotically.

from (2) that

$$3\alpha_1^2+4.5\alpha_1^3+6\alpha_1^4+8\alpha_1^5=4.5 \;\; s.t. \alpha_1^j\geq 0,\; \sum_{j=1}^{5}\alpha_1^j=1 \tag{3}$$

where $\alpha_i^j=a_{ij}/d_i$. Equation (3) has a solution $\alpha_1^2=0.55, \alpha_1^3=0.1, \alpha_1^4=0.2$, and $\alpha_1^5=0.15$. By the aforementioned procedure, one obtains α_i^j for all i,j. According to the

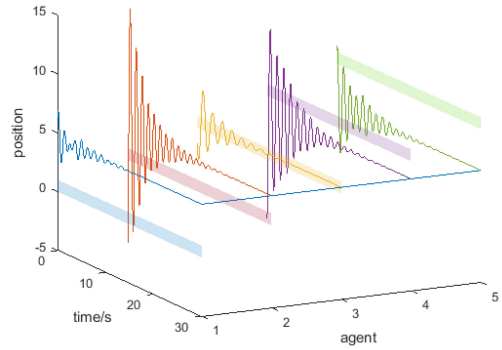


Fig. 9. Trajectory of agents with $\mathbf{H}_{ij}=\chi_j$ for $i,j=1,\ldots,N$. The constraint intervals are chosen to be the same as those in Fig. 8. The faded regions represent the constraint intervals. Even the constraint intervals admit an empty intersection, state consensus is asymptotically reached.

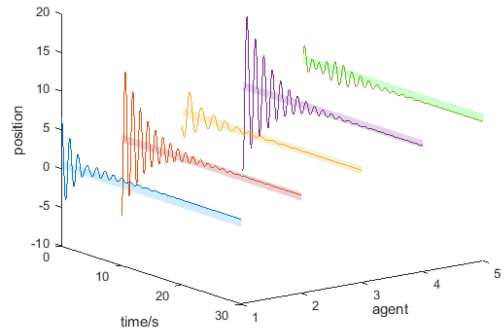


Fig. 10. Trajectory of agents with $\mathbf{H}_{ij}=\chi_i, i,j=1,\ldots,N$. The constraint intervals are chosen to be the same as those in Fig. 8. The faded regions represent the constraint intervals. Each agent finally motions into its own constraint interval. Therefore, state consensus is not realized.

obtained α_i^j, assign appropriate values to a_{ij} (the values of a_{ij} may not be unique). One thus has the following adjacency matrix of $\mathcal{G}$:

$$\mathsf{A}=\begin{bmatrix}0 & 11 & 2 & 4 & 3\\ 8 & 0 & 5 & 7 & 5\\ 1 & 4 & 0 & 4 & 1\\ 3 & 7 & 4 & 0 & 6\\ 3 & 3 & 4 & 10 & 0\end{bmatrix}.$$

It can be observed from Fig. 9 that consensus is asymptotically reached in spite of the empty intersection of the constraint intervals if $\mathbf{H}_{ij}=\chi_j, i,j\in\mathsf{V}$. However, Fig. 10 shows that each agent finally evolves into its own constraint interval and no consensus arises in the empty intersection case if $\mathbf{H}_{ij}=\chi_i, i,j=1,\ldots,N$.

VI. Conclusion

First, this article has been denoted to investigating the interval consensus problem for second-order MASs, where each agent imposes an interval constraint that functions to drive the agents into a favorable interval. Two different types of consensus frameworks, depending on whether the nodes themselves or their neighbors impose the interval constraints, have been

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considered. We have found, under both frameworks, that if the underlying graph is strongly connected and the intervals admit an empty intersection, then there exists a unique equilibrium point; while if the intersection is nonempty, then there exist multiple equilibrium points, all of which are consensus points. The convergence to a unique equilibrium point or consensus in a local manner has been analyzed via Lyapunov stability theory and matrix analysis techniques. We have also shown that if the distribution of the constraint intervals has certain properties, then global asymptotic convergence to the unique equilibrium point or to consensus can be guaranteed.

Second, agent interval consensus and neighbor interval consensus have significant difference. First, we need to introduce two different kinds of EPs, namely IEP and E-IEP, for stability analysis for the two frameworks. Second, conditions for the existence of an IEP and an E-IEP in the two frameworks are also quite different.

Finally, it is worth pointing out that even if the network is not strongly connected but contains a directed spanning tree, there may still exist a unique equilibrium point. For instance, if the constraint intervals admit an empty intersection for those nodes in a closed strong component, then it can be shown via the arguments developed in this article that there exists a unique equilibrium point. Further study on this issue is left for future works.

APPENDIX A
PRELIMINARY LEMMAS

We present here three lemmas that are used in the following proofs. Lemma 1 gives the square root of a complex number.

Lemma 1: Let $z = c + d \cdot \jmath$, where c and $d \neq 0$ are real numbers, and $\jmath$ is the imaginary unit. The square roots of z are $\pm(a + b \cdot \jmath)$ with $a = \sqrt{\frac{c+\sqrt{c^2+d^2}}{2}}$ and $b = \frac{d}{|d|}\sqrt{\frac{-c+\sqrt{c^2+d^2}}{2}}$, where $a > 0$ and b has the same sign as d.

We characterize one property of $\chi_m(\cdot)$ in the next lemma.

Lemma 2: Given any $x, y \in \mathbb{R}$ such that $x \neq y$, there holds that $|\chi_m(x) - \chi_m(y)| \leq |x - y|$ for any m. The equality holds if and only if $x, y \in \mathcal{I}_m$.

Proof: The proof is straightforward and omitted for brevity.

The following lemma shows that the position and velocity states of MAS (1) are bounded.

Lemma 3 (Bounded System State): Consider system (1). Given any initial values $\mathbf{x}_i(0)$ and $\mathbf{v}_i(0)$, $\mathbf{x}_i(t)$ and $\mathbf{v}_i(t)$ are bounded for all $i = 1, \ldots, N$.

Proof: We only prove the case that $\mathbf{H}_{ij} = \chi_j, i, j = 1, \ldots, N$. The other case can be proved in a similar way. Let $\mathbf{z}_i = [\mathbf{x}_i, \mathbf{v}_i]^\top$. The evolution of $\mathbf{z}_i$ is governed by

$$\dot{\mathbf{z}}_i = \underbrace{\begin{bmatrix} 0 & 1 \\ -d_i & -k \end{bmatrix}}_{\mathbf{A}_i} \mathbf{z}_i + \begin{bmatrix} 0 \\ s_i(t) \end{bmatrix}$$

where $d_i = \sum_{j\in\mathcal{N}_i} a_{ij}$ and $s_i(t) = \sum_{j\in\mathcal{N}_i} a_{ij}\chi_j(\mathbf{x}_j(t))$, which, apparently, is bounded. A simple calculation gives the eigenvalues of $\mathbf{A}_i$, denoted by $\lambda_1 = \frac{-k+\sqrt{k^2-4d_i}}{2}$ and $\lambda_2 = \frac{-k-\sqrt{k^2-4d_i}}{2}$, respectively.

Suppose that $d_i > 0$ for all $i \in \mathsf{V}$. Consequently, λ_1 and λ_2 both have negative real parts since $k > 0$, i.e., $\mathbf{A}_i$ is a Hurwitz matrix. Note that $\mathbf{z}_i(t)$ is expressible as

$$\mathbf{z}_i(t) = \exp\{\mathbf{A}_i t\}\,\mathbf{z}_i(0) + \int_0^t \exp\{\mathbf{A}_i(t-\tau)\}\begin{bmatrix} 0 \\ s_i(\tau) \end{bmatrix} \mathrm{d}\tau.$$

Since $s_i(t)$ is bounded and $\mathbf{A}_i$ is Hurwitz, $\|\mathbf{z}_i(t)\|$ is bounded, as desired. Suppose that there exists one node, say node i, having $d_i = 0$. Then, node i has no neighbor. As a consequence, $s_i(t) \equiv 0$ and $\mathbf{A}_i$ is marginally stable, i.e., the eigenvalues of $\mathbf{A}_i$ have nonpositive real parts and those lying on imaginary axis are semisimple. These facts directly give that $\|\mathbf{z}_i(t)\|$ is bounded.

APPENDIX B
PROOF OF THEOREM 1

We first characterize the equilibrium point for the nonempty intersection case in Appendix B-A. Then, we prove the uniqueness and existence of an equilibrium point for the empty intersection case in Appendix B-B and Appendix B-C, respectively.

A. Equilibrium Points in Nonempty Intersection Case

The proof of Theorem 1 makes use of Lemma 4 and its corollary, which characterize the positions of all agents if they have reached the equilibrium point.

Lemma 4: Let $[\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*]^\top$ be the solution of the system of nonlinear equations (2) with $\mathbf{H}_{ij} = \chi_j, i, j = 1, \ldots, N$, $p_* = \max_j\{p_j\}$, and $q_* = \min_j\{q_j\}$. Assume that $\mathcal{G}$ is strongly connected. If $p_* \leq q_*$, then $p_* \leq \mathbf{x}_i^* \leq q_*$ for all i, whereas if $q_* \leq p_*$, then $q_* \leq \mathbf{x}_i^* \leq p_*$ for all i.

Corollary 1: Let $[\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*]^\top$ be the solution of (2) with $\mathbf{H}_{ij} = \chi_j, i, j = 1, \ldots, N$, and $\mathcal{G}$ be strongly connected. There holds the following statements.

i) If $\cap_{j\in\mathsf{V}}\mathcal{I}_j \neq \emptyset$, then $\mathbf{x}_1^* = \cdots = \mathbf{x}_N^* \in \cap_{j\in\mathsf{V}}\mathcal{I}_j$.

ii) If $\mathbf{x}_i^* \in \mathcal{I}_i$ for all i, then $\cap_{j\in\mathsf{V}}\mathcal{I}_j \neq \emptyset$.

Proof of Lemma 4: We divide the proof into the following two cases.

Case I: $\mathbf{x}_i^* = \mathbf{x}_j^*$ for all $i \neq j$.

i) Let $p_* \leq q_*$. Suppose otherwise that $\mathbf{x}_i^* > q_*$ for all i. Then, $\mathbf{x}_i^* > q_k$ for some s satisfying $s \in \arg\min_j\{q_j\}$. Suppose s is a neighbor of ℓ, i.e., $(s, \ell) \in \mathsf{E}$. Then

$$\sum_{j\in\mathcal{N}_\ell} a_{\ell j}\left(\chi_j\left(\mathbf{x}_j^*\right) - \mathbf{x}_\ell^*\right) \leq a_{\ell s}\left(\chi_s\left(\mathbf{x}_s^*\right) - \mathbf{x}_\ell^*\right) < 0$$

where the first inequality holds due to the fact $\mathbf{x}_i^* > q_* \geq p_*$, which gives $\chi_i(\mathbf{x}_i^*) - \mathbf{x}_i^* \leq 0$, and the second inequality arises from $\chi_s(\mathbf{x}_s^*) \leq q_s < \mathbf{x}_i^* = \mathbf{x}_\ell^*$. This incurs a contradiction against the premise that $[\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*]^\top$ is a solution of (2). Therefore, $\mathbf{x}_i^* \leq q_*$ for all i. The proof concerning $\mathbf{x}_i \geq p_*$ for all i is almost the same as that for $\mathbf{x}_i \leq q_*$ for all i and is omitted.

ii) Let $q_* < p_*$. The analysis can be completed similarly and is omitted for brevity.

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Case II: $\exists i, j, \mathbf{x}_i^* \neq \mathbf{x}_j^*$.

i) Consider $p_* \leq q_*$. We first show $\mathbf{x}_j^* \leq q_*$ for all $j \in \mathsf{V}$. Suppose otherwise that there exists a nonempty index set Υ such that if $j \in \Upsilon$, then $\mathbf{x}_j^* > q_*$. Without loss of generality, let $\mathbf{x}_s^* = \max_{j\in\Upsilon} \mathbf{x}_j^*$. This s can be chosen in such a way that it is a neighbor of a node, say ℓ, that satisfies $\mathbf{x}_\ell^* < \mathbf{x}_s^*$. Such a choice is possible. Otherwise, it will be concluded that $\mathbf{x}_j^* = \mathbf{x}_s^* > q_*$ for any $j \in \mathsf{V}$ by the strong connectivity of $\mathcal{G}$. Now, we consider the term $\chi_v(\mathbf{x}_v^*) - \mathbf{x}_s^*$ for node v, a neighbor of s. Two cases arise, which are as follows.
   a) $\chi_v(\mathbf{x}_v^*) \leq \mathbf{x}_v^*$. Since $\mathbf{x}_s^* = \max_{j\in\Upsilon} \mathbf{x}_j^* \geq \mathbf{x}_v^*$, it is clear that $\chi_v(\mathbf{x}_v^*) - \mathbf{x}_s^* \leq \mathbf{x}_v^* - \mathbf{x}_s^* \leq 0$. Note that there exists one neighbor ℓ of s such that $\mathbf{x}_\ell^* < \mathbf{x}_s^*$. For this node, $\chi_\ell(\mathbf{x}_\ell^*) - \mathbf{x}_s^* < 0$.
   b) $\chi_v(\mathbf{x}_v^*) > \mathbf{x}_v^*$. This holds only if $\chi_v(\mathbf{x}_v^*) = p_v$. According to $p_* \leq q_*$, one has $p_v \leq p_* \leq q_* < \mathbf{x}_s^*$, then $\chi_v(\mathbf{x}_v^*) - \mathbf{x}_s^* = p_v - \mathbf{x}_s^* < 0$.

   Consequently, if there exists s such that $\mathbf{x}_s^* = \max_{j\in\Upsilon} \mathbf{x}_j^* > q_*$, then $\sum_{j\in\mathcal{N}_s} a_{sj}(\chi_j(\mathbf{x}_j^*) - \mathbf{x}_s^*) < 0$. Contradiction arises. The proof concerning $\mathbf{x}_j \geq p_*$ for all $j \in \mathsf{V}$ is almost the same as that for $\mathbf{x}_j \leq q_*$ for all $j \in \mathsf{V}$ and is omitted.

(ii) Consider $p_* > q_*$. The proof follows the idea in *Case II*(i). We show $\mathbf{x}_i^* \leq p_*$ for all $i \in \mathsf{V}$. Suppose otherwise that there exists j such that $\mathbf{x}_j^* > p_*$. We define Υ, s, and ℓ as in *Case II*(i), except that q_* is replaced by p_*. Now, we consider the term $\chi_v(\mathbf{x}_v^*) - \mathbf{x}_s^*$ for v, the neighbor of s. Two cases also arise, which are as follows.
   1) $\chi_v(\mathbf{x}_v^*) \leq \mathbf{x}_v^*$. By following the arguments in *Case II*(i), one has $\chi_v(\mathbf{x}_v^*) - \mathbf{x}_s^* \leq \mathbf{x}_v^* - \mathbf{x}_s^* \leq 0$. Moreover, there exists one neighbor ℓ of s such that $\chi_\ell(\mathbf{x}_\ell^*) - \mathbf{x}_s^* < 0$.
   2) $\chi_v(\mathbf{x}_v^*) > \mathbf{x}_v^*$. As a result, $\chi_v(\mathbf{x}_v^*) = p_v$. Since $p_* \geq p_v$, $\chi_v(\mathbf{x}_v^*) - \mathbf{x}_s^* = p_v - \mathbf{x}_s^* \leq p_* - \mathbf{x}_s^* < 0$.

   Therefore, if there exists s such that $\mathbf{x}_k^* = \max_{j\in\Upsilon} \mathbf{x}_j^* > p_*$, then $\sum_{j\in\mathcal{N}_s} a_{sj}(\chi_j(\mathbf{x}_j^*) - \mathbf{x}_s^*) < 0$ – a contradiction. The other part of the proof for p_* is the same as that for q_*.

Proof of Corollary 1: (i) Since $\cap_{j\in\mathsf{V}}\mathcal{I}_j \neq \emptyset$, one has $p_* \leq q_*$. Then, by the strong connectivity of $\mathcal{G}$ and Lemma 4, $\mathbf{x}_i^* \in [p_*, q_*]$ for all $i \in \mathsf{V}$. As a result, $\chi_j(\mathbf{x}_j^*) = \mathbf{x}_j^*$ for all $i \in \mathsf{V}$. This implies that

$$\sum_{j\in\mathcal{N}_i} a_{ij}\left(\mathbf{x}_j^* - \mathbf{x}_i^*\right) = 0,\ i = 1, \ldots, N \tag{4}$$

which, according to strong connectivity of $\mathcal{G}$ [4], yields $\mathbf{x}_1^* = \cdots = \mathbf{x}_N^* \in [p_*, q_*]$, as desired.

(ii) Due to $\mathbf{x}_i^* \in \mathcal{I}_i$, the system of nonlinear equations (2) turns to be in the form of (4). Then, the strong connectivity of $\mathcal{G}$ implies that $\mathbf{x}_1^* = \cdots = \mathbf{x}_N^*$ [4], which in turn gives that $\cap_{j\in\mathsf{V}}\mathcal{I}_j$ contains at least one point and is not empty.

Proof of Theorem 1(ii): By Corollary 1(i), $\mathbf{x}_1^* = \cdots = \mathbf{x}_N^* \in [p_*, q_*]$.

Next, with Lemma 4 and its corollary, we further prove Theorem 1 for the case that the constraint intervals admit an empty intersection $\cap_{j=1}^N \mathcal{I}_j = \emptyset$. To this end, it suffices to show the existence and uniqueness of a solution of (2).

B. Uniqueness of the Equilibrium Point in Empty Intersection Case

In this part, we show that if a solution of (2) exists, then it is unique. Suppose otherwise that the system of nonlinear equations (2) has two solutions, namely $\mathbf{x}^* = [\mathbf{x}_1^*, \ldots, \mathbf{x}_N^*]^\top$ and $\mathbf{y}^* = [\mathbf{y}_1^*, \ldots, \mathbf{y}_N^*]^\top$. It follows, for all $i = 1, \ldots, N$, that

$$\mathbf{x}_i^* = \sum\nolimits_{j\in\mathcal{N}_i} \alpha_i^j \chi_j\left(\mathbf{x}_j^*\right) \ \text{ and } \ \mathbf{y}_i^* = \sum\nolimits_{j\in\mathcal{N}_i} \alpha_i^j \chi_j\left(\mathbf{y}_j^*\right) \tag{5}$$

where $\alpha_i^j = a_{ij} / \sum_{j\in\mathcal{N}_i} a_{ij}$. Define $\beta_i = \mathbf{y}_i^* - \mathbf{x}_i^*$. Then, one has for all i that

$$\mathbf{y}_i^* - \beta_i = \sum_{j\in\mathcal{N}_i} \alpha_i^j \chi_j\left(\mathbf{y}_j^* - \beta_j\right). \tag{6}$$

Without loss of generality, let $\mathbf{x}_1^* < \mathbf{y}_1^*$. Therefore, $\beta_1 = \mathbf{y}_1^* - \mathbf{x}_1^* > 0$. From (5) and (6), one has for the first agent that

$$\sum_{j\in\mathcal{N}_1} \alpha_1^j \left[\chi_j\left(\mathbf{y}_j^*\right) - \beta_1 - \chi_j\left(\mathbf{y}_j^* - \beta_j\right)\right] = 0. \tag{7}$$

Hence, there exists $p_1 \in \mathcal{N}_1$ such that $\chi_{p_1}(\mathbf{y}_{p_1}^*) - \chi_{p_1}(\mathbf{y}_{p_1}^* - \beta_{p_1}) \geq \beta_1$, which, by Lemma 2, immediately gives that $\beta_{p_1} \geq \beta_1$. Now, we have

$$\mathbf{y}_{p_1}^* - \beta_{p_1} = \sum_{j\in\mathcal{N}_{p_1}} \alpha_{p_1}^j \chi_j\left(\mathbf{y}_j^* - \beta_j\right) \tag{8}$$

and $\beta_{p_1} = \mathbf{y}_{p_1}^* - \mathbf{x}_{p_1}^* \geq \beta_1 > 0$. Suppose that $\mathcal{N}_{p_1} \setminus \{1\}$ is not empty. By similar arguments, it is obtained that there exists $p_2 \in \mathcal{N}_{p_1}$ and $p_2 \neq 1$ such that $\beta_{p_2} \geq \beta_{p_1}$. Otherwise, suppose that $\chi_j(\mathbf{y}_j^*) - \chi_j(\mathbf{y}_j^* - \beta_j) \geq \beta_{p_1}$, $j \in \mathcal{N}_{p_1}$, holds only for $j = 1$. Then, it is obtained from Lemma 2 and $\beta_{p_1} \geq \beta_1$ that $\chi_1(\mathbf{y}_1^*) - \chi_1(\mathbf{y}_1^* - \beta_1) = \beta_{p_1} = \beta_1$, which implies that (8) does not hold. If $\mathcal{N}_{p_1} = \{1\}$, then $\chi_1(\mathbf{y}_1^*) - \chi_1(\mathbf{y}_1^* - \beta_1) = \beta_{p_1} = \beta_1$. As a result, p_2 can be found from $\mathcal{N}_1$, which satisfies that $\mathcal{N}_1 \setminus \{p_1\} \neq \emptyset$, as discussed for the case that $\mathcal{N}_{p_1} \setminus \{1\}$ is not empty. By continuing this procedure and strong connectivity of $\mathcal{G}$, there holds that $\beta_{p_j} \geq \beta_{p_i} > 0$, $j \geq 1$, $0 \leq i \leq j-1$, where $1 \leq p_s \leq N$ for $s = 0, \ldots, N-1$ and $p_0 = 1$.

Now, consider

$$\mathbf{y}_{p_{N-1}}^* - \beta_{p_{N-1}} = \sum_{j\in\mathcal{N}_{p_{N-1}}} \alpha_{p_{N-1}}^j \chi_j\left(\mathbf{y}_j^* - \beta_j\right). \tag{9}$$

Bearing (5) in mind, (9) can be equivalently written as

$$\sum_{j\in\mathcal{N}_{p_{N-1}}} \alpha_{p_{N-1}}^j \left[\chi_j\left(\mathbf{y}_j^*\right) - \beta_{p_{N-1}} - \chi_j\left(\mathbf{y}_j^* - \beta_j\right)\right] = 0.$$

By Lemma 2, it is obtained for all j that

$$\chi_j\left(\mathbf{y}_j^*\right) - \chi_j\left(\mathbf{y}_j^* - \beta_j\right) \leq \beta_j \leq \beta_{p_{N-1}}. \tag{10}$$

Hence, equality (9) is true if only $\beta_{p_{N-1}} = \beta_{p_j}$ for all $p_j \in \mathcal{N}_{p_{N-1}}$. Since $\mathcal{G}$ is strongly connected, one further obtains that $\beta_{p_0} = \cdots = \beta_{p_{N-1}} = \beta$ for some β. However, since $\cap_{j=1}^N \mathcal{I}_j = \emptyset$, for some $s \in \mathsf{V}$, it follows from Corollary 1(ii) that $\mathbf{y}_s^* \notin \mathcal{I}_s$.

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Therefore, according to Lemma 2, $\chi_s(\mathbf{y}_s^*) - \chi_s(\mathbf{y}_s^* - \beta) < \beta$. Then, if $s \in \mathcal{N}_i$ for some i, one has from (10) that

$$\mathbf{y}_i^* - \beta = \sum_{j\in\mathcal{N}_i} \alpha_i^j \left(\chi_j\left(\mathbf{y}_j^*\right) - \beta\right) < \sum_{j\in\mathcal{N}_i} \alpha_i^j \chi_j\left(\mathbf{y}_j^* - \beta\right)$$

a contradiction against (6). Consequently, if solution of (2) exists, then it is unique. ■

C. Existence of an Equilibrium Point in Empty Intersection Case

In this part, we prove that the solution of (2) exists. To complete the proof, we resort to mathematical induction. Given any $\mathbf{x}_i, 1 \le i \le N$, suppose that $\mathbf{x}_1 > \sum_{j\in\mathcal{N}_1} \alpha_1^j \chi_j(\mathbf{x}_j)$. Fix $\mathbf{x}_i, i = 2, \ldots, N$. Then, there exists $r > 0$ such that $\mathbf{x}_1 - r = \sum_{j\in\mathcal{N}_1} \alpha_1^j \chi_j(\mathbf{x}_j)$.

Now, assume that given any $k \ge 2$ and any $\mathbf{x}_i, 1 \le i \le N$, there exist r_i, $i < k$, such that

$$\mathbf{x}_i^1 = \sum\nolimits_{j\in\mathcal{N}_i} \alpha_i^j \chi_j\left(\mathbf{x}_j^1\right) \qquad \forall i < k \tag{11}$$

where $\mathbf{x}_i^1 = \mathbf{x}_i - r_i, i < k$ and $\mathbf{x}_i^1 = \mathbf{x}_i, i \ge k$. In what follows, we fix $\mathbf{x}_i^1, i \ge k+1$. By the induction premise, for any $\mathbf{x}_k^1$, there exist $\mathbf{x}_i^1, i < k$, such that (11) holds. Our aim is to prove the existence of $s_i, i \le k$, such that there holds $\mathbf{x}_i^2 = \sum_{j\in\mathcal{N}_i} \alpha_i^j \chi_j(\mathbf{x}_j^2)$ for $i \le k$, where

$$\mathbf{x}_i^2 = \begin{cases} \mathbf{x}_i^1 - s_i, & i \le k \\ \mathbf{x}_i^1, & i \ge k+1. \end{cases}$$

For simplicity, define $s_i = 0, i \ge k+1$. Then, $\mathbf{x}_i^2 = \mathbf{x}_i^1 - s_i$ for all $i = 1, \ldots, N$.

Note that given any s_k, there exist $s_i, i \le k-1$, such that (11) holds for $\mathbf{x}_i^2, 1 \le i \le k-1$. If such s_i are unique, then s_i are functions of s_k, i.e., $s_i = g_i(s_k), i \le k-1$. We claim that g_i is a *continuous* function of s_k. Next, we show that: 1) s_i are functions of s_k, i.e., $s_i = g_i(s_k)$, and 2) $g_i(s_k)$ are continuous for $i = 1, \ldots, k-1$.

1) In this step, we prove that given any s_k, the values s_i, $i < k$, can be uniquely determined such that (11) holds for $\mathbf{x}_i^2, i \le k-1$. Define $\Upsilon_s = \{s+1, \ldots, N\}$ and $\bar{\Upsilon}_s = \{1, \ldots, s\}$ for any $1 \le s \le N$. For the aforementioned aim, it suffices to show that for $\bar{\mathbf{x}} = [\bar{\mathbf{x}}_1, \ldots, \bar{\mathbf{x}}_s, \mathbf{x}_{s+1}, \ldots, \mathbf{x}_N]^\top$ and $\hat{\mathbf{x}} = [\hat{\mathbf{x}}_1, \ldots, \hat{\mathbf{x}}_s, \mathbf{x}_{s+1}, \ldots, \mathbf{x}_N]^\top$ where $1 \le s \le N$, if

$$\bar{\mathbf{x}}_i = \sum_{j\in\mathcal{N}_i\setminus\Upsilon_s} \alpha_i^j \chi_j\left(\bar{\mathbf{x}}_j\right) + \sum_{j\in\Upsilon_s} \alpha_i^j \chi_j\left(\mathbf{x}_j\right)$$

and

$$\hat{\mathbf{x}}_i = \sum_{j\in\mathcal{N}_i\setminus\Upsilon_s} \alpha_i^j \chi_j\left(\hat{\mathbf{x}}_j\right) + \sum_{j\in\Upsilon_s} \alpha_i^j \chi_j\left(\mathbf{x}_j\right)$$

hold for all $i = 1, \ldots, s$, then one has $\bar{\mathbf{x}} = \hat{\mathbf{x}}$.

Define $\beta_i = \bar{\mathbf{x}}_i - \hat{\mathbf{x}}_i$, $1 \le i \le s$. Suppose otherwise that $\bar{\mathbf{x}} \ne \hat{\mathbf{x}}$. Let $\bar{\mathbf{x}}_1 - \hat{\mathbf{x}}_1 = \beta_1 > 0$ without loss of generality. By following the arguments developed for the proof in Appendix B-B, one obtains that

$\beta_{p_s} \ge \beta_{p_{s-1}} \ge \cdots \ge \beta_{p_1} > 0$, where $1 \le p_j \le s$ for all $j \le s$ and $\beta_{p_1} = \beta_1$.

Suppose that $\beta_{p_s} = \bar{\mathbf{x}}_p - \hat{\mathbf{x}}_p$ for some $1 \le p \le s$. Then, one has

$$\bar{\mathbf{x}}_p - \hat{\mathbf{x}}_p = \sum_{j\in\mathcal{N}_p\setminus\Upsilon_s} \alpha_p^j \left[\chi_j\left(\bar{\mathbf{x}}_j\right) - \chi_j\left(\hat{\mathbf{x}}_j\right)\right] = \beta_{p_s}. \tag{12}$$

Note that $\sum_{j\in\mathcal{N}_p\setminus\Upsilon_s} \alpha_p^j \le 1$ and $0 \le \chi_j(\bar{\mathbf{x}}_j) - \chi_j(\hat{\mathbf{x}}_j) \le \bar{\mathbf{x}}_j - \hat{\mathbf{x}}_j \le \beta_{p_s}$ where $1 \le j \le s$. Equation (12) holds if only $\mathcal{N}_p \subset \bar{\Upsilon}_s$ and $\bar{\mathbf{x}}_j - \hat{\mathbf{x}}_j = \beta_{p_s}$ for all $j \in \mathcal{N}_p$. By continuing the aforementioned procedure and also following the arguments developed for the proof in Appendix B-B, one obtains from the strong connectivity of $\mathcal{G}$ that

$$\bar{\mathbf{x}}_i - \hat{\mathbf{x}}_i = \beta_{p_s} \quad \forall i \in \bar{\Upsilon}_s. \tag{13}$$

However, since $\mathcal{G}$ is strongly connected, there exists $w \in \bar{\Upsilon}_s$ such that $\mathcal{N}_w \cap \Upsilon_s \ne \emptyset$. Consequently, $\bar{\mathbf{x}}_w - \hat{\mathbf{x}}_w = \sum_{j\in\mathcal{N}_w\setminus\Upsilon_s} \alpha_w^j \left[\chi_j(\bar{\mathbf{x}}_j) - \chi_j(\hat{\mathbf{x}}_j)\right] < \beta_{p_s}$, where $\sum_{j\in\mathcal{N}_w\setminus\Upsilon_s} \alpha_w^j < 1$. This contradicts (13) and finishes the first step.

2) In this step, we prove that $g_i(s_k)$ are continuous functions of s_k for $i \le k-1$. Recall that given any s_k, we have $\mathbf{x}_i^2 = \sum_{j\in\mathcal{N}_i} \alpha_i^j \chi_j(\mathbf{x}_j^2)$ for $i < k$, where $\mathbf{x}_i^2 = \mathbf{x}_i^1 - s_i, i = 1, \ldots, N$.

At no loss of generality, let $s_k > 0$. We first show that $0 \le s_i \le s_k$ for $i = 1, \ldots, k-1$. Suppose that there exists p_1 such that $s_{p_1} < 0$, where $1 \le p_1 \le k-1$. It follows from the induction premise that $\mathbf{x}_{p_1}^1 - s_{p_1} = \sum_{j\in\mathcal{N}_{p_1}} \alpha_{p_1}^j \chi_j(\mathbf{x}_j^1) - s_{p_1} = \sum_{j\in\mathcal{N}_{p_1}} \alpha_{p_1}^j \chi_j(\mathbf{x}_j^1 - s_j)$. In view of the fact that $\chi_j(\mathbf{x}_j^1) - \chi_j(\mathbf{x}_j^1 - s_j) \ge 0$ for $s_j \ge 0$, there exists at least one neighbor p_2 of agent p_1 such that $s_{p_2} < 0$, where $1 \le p_2 \le k-1$ due to $s_j = 0$, $j \ge k+1$. In particular, this agent p_2 can be chosen to satisfy that $s_{p_2} \le s_{p_1}$ as follows:

$$\begin{aligned} 0 > s_{p_1} &= \sum_{j\in\mathcal{N}_{p_1}} \alpha_{p_1}^j \left[\chi_j\left(\mathbf{x}_j^1\right) - \chi_j\left(\mathbf{x}_j^1 - s_j\right)\right] \\ &\ge \underbrace{\chi_{p_2}\left(\mathbf{x}_{p_2}^1\right) - \chi_{p_2}\left(\mathbf{x}_{p_2}^1 - s_{p_2}\right)}_{p_2\in\arg\min_{j\in\mathcal{N}_{p_1}}\{\chi_j(\mathbf{x}_j^1) - \chi_j(\mathbf{x}_j^1 - s_j)\}} \ge s_{p_2}. \end{aligned}$$

By continuing the aforementioned procedures and following the arguments developed for the proof in Appendix B-B, one has $s_{p_i}, 1 \le i \le k-1$, which satisfy that

$$s_{p_{k-1}} \le s_{p_{k-2}} \le \cdots \le s_{p_1} < 0 < s_k \tag{14}$$

and $1 \le p_i \le k-1$. We show in what follows that (14) eventually leads to a contradiction. Since $\mathcal{G}$ is strongly connected, p_{k-1} has at least one neighbor and

$$\begin{aligned} s_{p_{k-1}} &= \sum_{j\in\mathcal{N}_{p_{k-1}}} \alpha_{p_{k-1}}^j \chi_j\left(\mathbf{x}_j^1\right) - \sum_{j\in\mathcal{N}_{p_{k-1}}} \alpha_{p_{k-1}}^j \chi_j\left(\mathbf{x}_j^1 - s_j\right) \\ &\ge \min_{j\in\mathcal{N}_{p_{k-1}}} \left\{\chi_j\left(\mathbf{x}_j^1\right) - \chi_j\left(\mathbf{x}_j^1 - s_j\right)\right\} \ge \min_{j\in\mathcal{N}_{p_{k-1}}} s_j. \end{aligned}$$

The last two equalities hold if only $\mathcal{N}_{p_{k-1}} \subset \bar{\Upsilon}_{k-1}$ and $s_m = s_n$ for $m, n \in \mathcal{N}_{p_{k-1}}$. Since $s_j = 0$ for $j \ge k+1$, it follows from the strong connectivity of $\mathcal{G}$ that (14) is true only if

$$s_m = s_n < 0 \qquad \forall\, m, n \in \bar{\Upsilon}_{k-1}. \tag{15}$$

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However, by the strong connectivity of $\mathcal{G}$, there exists at least one node, say $p \in \bar{\Upsilon}_{k-1}$, having a neighbor from the set $\mathsf{V} \setminus \bar{\Upsilon}_{k-1}$. Hence

$$s_p = \sum\nolimits_{j\in\mathcal{N}_p\setminus\Upsilon_{k-1}} \alpha_p^j \left[\chi_j\left(\mathbf{x}_j^1\right) - \chi_j\left(\mathbf{x}_j^1 - s_j\right)\right]$$
$$\geq \left(\sum\nolimits_{j\in\mathcal{N}_p\setminus\Upsilon_{k-1}} \alpha_p^j\right) \min_{j\in\mathcal{N}_p\setminus\Upsilon_{k-1}} s_j > \min\nolimits_{j\in\mathcal{N}_p\setminus\Upsilon_{k-1}} s_j$$

where the last inequality holds due to $\sum_{j\in\mathcal{N}_p\setminus\Upsilon_{k-1}} \alpha_p^j < 1$. A contradiction against (15) arises. This gives that $s_p \geq 0, p = 1,\ldots,k-1$. The proof for $s_i \leq s_k$, $i \leq k-1$, can be completed in a similar way and is omitted for simplicity.

The aforementioned discussions give that $|s_i| \leq |s_k|$ for $i \leq k-1$. Now, given any $\epsilon > 0$ and z_1, z_2 satisfying that $|z_1 - z_2| \leq \epsilon$. Letting $\mathbf{x}_k^1 = z_1$ and $s_k = z_1 - z_2$, it is ready to have $|g_i(z_1) - g_i(z_2)| \leq |s_k| \leq \epsilon$. This implies the continuity of $g_i(s_k)$ for $i = 1,\ldots,k-1$.

We have shown that $g_i(s_k)$ are continuous functions for $i \leq k-1$. Now, define $f(s_k)$: $f(s_k) = \mathbf{x}_k^1 - s_k - \sum_{j\in\mathcal{N}_k} \alpha_k^j \chi_j(\mathbf{x}_j^1 - g_j(s_k))$, where $g_j(s_k) = 0$ for $j \geq k+1$. Note that the continuity of $g_i(s_k)$ guarantees the continuity of $f(s_k)$. If there exists a real number s_k such that $f(s_k) = 0$, then evidently, with $s_i = g_i(s_k)$, one has $\mathbf{x}_i^1 - s_i = \sum_{j\in\mathcal{N}_i} \alpha_i^j \chi_j(\mathbf{x}_j^1 - s_j)$ for $1 \leq i \leq k$. Next, we prove the existence of s_k. If $f(0) = 0$, then we have found $\mathbf{x}_i^2 = \mathbf{x}_i^1, i \leq k$. Suppose at no loss of generality that $f(0) > 0$. Note also that $\lim_{s_k\to+\infty} f(s_k) = -\infty$ since $\chi_i(\cdot)$ is bounded. Therefore, the existence of s_k to ensure $f(s_k) = 0$ follows from the continuity of $f(s_k)$. This completes the proof for the existence of a solution of (2). ■

APPENDIX C
PROOF OF THEOREM 2

A. Technical Lemma

In the proof of Theorem 2, we make use of the following lemma.

Lemma 5: Consider the agent interval consensus. Let $\mathbf{x} = [\mathbf{x}_1,\ldots,\mathbf{x}_N]^\top \in \mathbb{R}^N$ be fixed. Assume that $\mathcal{G}$ is strongly connected. If there exists $i \in \mathsf{V}$ such that $\chi_i(\mathbf{x}_i) \neq \mathbf{x}_i$, then the system of nonlinear equations (2) can be written as $-\hat{\mathbf{L}}_1\mathbf{x} + \mathbf{b} = \mathbf{0}$, where $\hat{\mathbf{L}}_1$ has eigenvalues with positive real parts and $\mathbf{b}$ is some constant vector.

Proof: By the conditions in the lemma statement, there exists $i \in \mathsf{V}$ such that $\chi_i(\mathbf{x}_i) \neq \mathbf{x}_i$. Define $\Upsilon = \{k \in \mathsf{V} \,|\, \chi_k(\mathbf{x}_k) \neq \mathbf{x}_k\}$, which is not empty. Assume that $\chi_k(\mathbf{x}_k) = y_k \in \{p_k, q_k\}$ for $k \in \Upsilon$. Then, for $1 \leq i \leq N$, there holds that $\sum_{j\in\mathcal{N}_i\setminus\Upsilon} a_{ij}(\mathbf{x}_j(t) - \mathbf{x}_i(t)) + \sum_{j\in\Upsilon} a_{ij}(y_j - \mathbf{x}_i(t)) = 0$. Write the aforementioned equations into a compact form as

$$-\mathbf{L}_1\mathbf{x} - \operatorname{diag}\left\{\sum_{j\in\Upsilon} a_{1j},\ldots,\sum_{j\in\Upsilon} a_{Nj}\right\}\mathbf{x} + \mathbf{b} = \mathbf{0}$$

where $\mathbf{b} = [\mathbf{b}_1,\ldots,\mathbf{b}_N]^\top$, $\mathbf{b}_i = \sum_{j\in\Upsilon} a_{ij}y_j$, and $\mathbf{L}_1$ is defined in Section III. Note that $\mathbf{L}_1$ can be transformed into the Frobenius normal form [35]. For brevity, we simply assume that $\mathbf{L}_1$ takes Frobenius normal form

$$\mathbf{L}_1 = \begin{bmatrix} \mathbf{L}_1^{11} & & & & & \\ \vdots & \ddots & & & & \\ \mathbf{0} & \cdots & \mathbf{L}_1^{\psi\psi} & & & \\ \star & \cdots & \star & \mathbf{L}_1^{\psi+1,\psi+1} & & \\ \vdots & \cdots & \ddots & \star & \ddots & \\ \star & \cdots & \cdots & \cdots & \star & \mathbf{L}_1^{\phi,\phi} \end{bmatrix}$$

where $\mathbf{L}_1^{ii}$ is a Laplacian matrix of a strongly connected graph $\mathcal{G}_1^{ii}$ for $i = 1,\ldots,\psi$. Note that, throughout the Appendix, we use $\star$ to represent terms that are compatible for algebraic operations but need not be stated explicitly. Since $\mathcal{G}$ is strongly connected, there exists at least one node, say k in $\mathcal{G}_1^{ii}$, such that $\sum_{j\in\Upsilon} a_{kj} > 0$ for $i = 1,\ldots,\psi$. It is therefore known from [35, Lemma A.8] that the matrix $-\hat{\mathbf{L}}_1 = -\mathbf{L}_1 - \operatorname{diag}\{\sum_{j\in\Upsilon} a_{1j},\ldots,\sum_{j\in\Upsilon} a_{Nj}\}$ is a Hurwitz matrix.

B. Proof of Theorem 2(i) and (ii)

We use Lyapunov's indirect method (see, e.g., [36]) to prove local stability of the IEP $\mathbf{z}^*$.

Since $\mathcal{I}_m \cap \mathcal{I}_n = \emptyset$ for some $m, n \in \mathsf{V}$, by Corollary 1(ii), there exists at least one node, say j, such that $\mathbf{x}_j^* \neq \chi_j(\mathbf{x}_j^*)$. According to Lemma 5, write $\sum_{j\in\mathcal{N}_i} a_{ij}(\chi_j(\mathbf{x}_j^*) - \mathbf{x}_i^*)$, $i = 1,\ldots,N$, as $-\hat{\mathbf{L}}_1\mathbf{x}^* + \mathbf{b}$, where $-\hat{\mathbf{L}}_1$ is Hurwitz, i.e., $\mathsf{Re}(\lambda_i(\hat{\mathbf{L}}_1)) > 0$ for any $i = 1,\ldots,N$. Rewrite (1) in a compact form around $\mathbf{z}^*$ as $\dot{\mathbf{z}} = \mathbf{f}(\mathbf{z})$ with $\mathbf{f}(\mathbf{z}) = \begin{bmatrix} \mathbf{0} & \mathbf{I}_N \\ -\hat{\mathbf{L}}_1 & -k\mathbf{I}_N \end{bmatrix}\mathbf{z} + \begin{bmatrix} \mathbf{0} \\ \mathbf{b} \end{bmatrix}$, where $\mathbf{b}$ is a constant vector near $\mathbf{z}^*$ as can be observed from the proof of Lemma 5. Then, linearizing $\mathbf{f}(\mathbf{z})$ at the IEP $\mathbf{z}^*$ yields a Jacobian matrix $\Gamma = \begin{bmatrix} \mathbf{0}_{N\times N} & \mathbf{I}_N \\ -\hat{\mathbf{L}}_1 & -k\mathbf{I}_N \end{bmatrix}$, whose spectrum determines whether $\mathbf{z}^*$ is locally stable.

Next, we calculate each eigenvalue λ of Γ, which requires to solve the equation: $\det(\lambda\mathbf{I} - \Gamma) = 0$. Note that

$$\det(\lambda\mathbf{I} - \Gamma) = \det\left(\lambda\mathbf{I} - \begin{bmatrix} \mathbf{0}_{N\times N} & \mathbf{I}_N \\ -\hat{\mathbf{L}}_1 & -k\mathbf{I}_N \end{bmatrix}\right)$$
$$= \det\left(\begin{bmatrix} \lambda\mathbf{I}_N & -\mathbf{I}_N \\ \hat{\mathbf{L}}_1 & (\lambda+k)\mathbf{I}_N \end{bmatrix}\right) = \det\left(\lambda(\lambda+k)\mathbf{I}_n + \hat{\mathbf{L}}_1\right).$$

Let $\mu_i = \mathsf{Re}(\mu_i) + j\mathsf{Im}(\mu_i)$ be an eigenvalue of $\hat{\mathbf{L}}_1$. Recall that μ_i has a positive real part, i.e., $\mathsf{Re}(\mu_i) > 0$. Since any eigenvalue λ of Γ satisfies the equation $\det(\lambda(\lambda+k)\mathbf{I}_n + \hat{\mathbf{L}}_1) = 0$, one has

$$\lambda(\lambda+k) + \mu_i = 0$$
$$\Longrightarrow \lambda = \frac{-k \pm \sqrt{k^2 - 4\mu_i}}{2}$$
$$= \frac{-k \pm \sqrt{k^2 - 4\mathsf{Re}(\mu_i) - 4j\mathsf{Im}(\mu_i)}}{2}.$$

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According to Lemma 1, the real part of $\sqrt{k^2-4\mathsf{Re}(\mu_i)-4\jmath\mathsf{Im}(\mu_i)}$ is $\pm\sigma(k)$ with $\sigma(k)=\sqrt{\frac{k^2-4\mathsf{Re}(\mu_i)+\sqrt{(k^2-4\mathsf{Re}(\mu_i))^2+(4\mathsf{Im}(\mu_i))^2}}{2}}>0$. If $\mathsf{Im}(\mu_i)=0$, then it is obvious that $-k\pm\sigma(k)$ have negative real parts. Next, we consider the case $\mathsf{Im}(\mu_i)\neq 0$ and figure out conditions for $-k+\sigma(k)<0$, since $-k-\sigma(k)<0$. Via simple calculations, one has

$$\begin{aligned}-k+\sigma(k)<0 &\Longleftrightarrow \frac{k^2-a+\sqrt{(k^2-a)^2+b^2}}{2}<k^2\\ &\Longleftrightarrow (k^2-a)^2+b^2<(k^2+a)^2\\ &\Longleftrightarrow \frac{b^2}{4a}<k^2 \Leftarrow 0<\sqrt{\frac{b^2}{4a}}<k\end{aligned}$$

where $a=4\mathsf{Re}(\mu_i)>0$ and $b^2=(4\mathsf{Im}(\mu_i))^2\neq 0$. Therefore, there exists a critical value of k, denoted by $k^*=\max_{i\in\{1,\ldots,N\}}|\mathsf{Im}(\mu_i)|/\sqrt{\mathsf{Re}(\mu_i)}$, such that if $k>k^*$, then Γ is a Hurwitz matrix, whereas if $k<k^*$, then there exists at least one eigenvalue of Γ having positive real part, which directly gives that the IEP is not stable [36].

In what follows, assume that Γ is Hurwitz. We aim to find a small vicinity $\mathscr{B}(\mathbf{z}^*,r_0)=\{\mathbf{z}\in\mathbb{R}^{2N}|\|\mathbf{z}-\mathbf{z}^*\|\leq r_0\}$ of the IEP $\mathbf{z}^*$ in which $\mathbf{x}_j\in(p_j,q_j)$ or $\mathbf{x}_j\in\mathbb{R}\setminus[p_j,q_j]$ for all $j\in\mathsf{V}$, and prove that if $\mathbf{z}_0\in\mathscr{B}(\mathbf{z}^*,r)\subset\mathscr{B}(\mathbf{z}^*,r_0)$ for some $r>0$, then $\mathbf{z}(t)\in\mathscr{B}(\mathbf{z}^*,r_0)$ for any t and $\mathbf{z}(t)\to\mathbf{z}^*$ as $t\to\infty$.

Let $\hat{\mathbf{z}}=\mathbf{z}-\mathbf{z}^*=[\mathbf{x}^\top,\mathbf{v}^\top]^\top-[(\mathbf{x}^*)^\top,(\mathbf{v}^*)^\top]^\top$, where $\mathbf{z}\in\mathscr{B}(\mathbf{z}^*,r)$. It can be obtained from system (1) that

$$\dot{\hat{\mathbf{z}}}=\mathbf{f}(\mathbf{z})-\mathbf{f}(\mathbf{z}^*)=\left.\frac{\partial\mathbf{f}}{\partial\mathbf{z}}\right|_{\mathbf{z}^*}\hat{\mathbf{z}}+\mathbf{g}(\hat{\mathbf{z}}) \tag{16}$$

where $\left.\frac{\partial\mathbf{f}}{\partial\mathbf{z}}\right|_{\mathbf{z}^*}=\Gamma=\begin{bmatrix}\mathbf{0}_{N\times N} & \mathbf{I}_N\\ -\hat{\mathbf{L}}_1 & -k\mathbf{I}_N\end{bmatrix}$ is a Hurwitz matrix. Moreover, $\mathbf{g}(\hat{\mathbf{z}})$ satisfies that $\|\mathbf{g}(\hat{\mathbf{z}})\|/\|\hat{\mathbf{z}}\|\to 0$ as $\|\hat{\mathbf{z}}\|\to 0$ [36]. Let $\mathbf{P}>\mathbf{0}$ be such that $\Gamma^\top\mathbf{P}+\mathbf{P}\Gamma<\mathbf{0}$. The existence of $\mathbf{P}$ is guaranteed by the fact that Γ is Hurwitz. Define the Lyapunov function candidate $V=\hat{\mathbf{z}}^\top\mathbf{P}\hat{\mathbf{z}}$, whose derivative along (16) is

$$\dot{V}=2\hat{\mathbf{z}}^\top\mathbf{P}\left(\Gamma\hat{\mathbf{z}}+\mathbf{g}(\hat{\mathbf{z}})\right)=\hat{\mathbf{z}}^\top\left(\Gamma^\top\mathbf{P}+\mathbf{P}\Gamma\right)\hat{\mathbf{z}}+2\hat{\mathbf{z}}^\top\mathbf{P}\mathbf{g}(\hat{\mathbf{z}}).$$

By the property of $\mathbf{g}(\hat{\mathbf{z}})$, given any $\gamma>0$, there exists $r(\gamma)>0$ such that $\|\mathbf{g}(\hat{\mathbf{z}})\|\leq\gamma\|\hat{\mathbf{z}}\|\quad\forall\|\hat{\mathbf{z}}\|\leq r(\gamma)$. Let $\lambda_{\max}(\Gamma^\top\mathbf{P}+\mathbf{P}\Gamma)+2\lambda_{\max}(\mathbf{P})\gamma_1=0$, which yields

$$\gamma_1=-\lambda_{\max}(\Gamma^\top\mathbf{P}+\mathbf{P}\Gamma)/(2\lambda_{\max}(\mathbf{P}))>0.$$

Let $r>0$ be defined by $r\sqrt{\lambda_{\max}(\mathbf{P})/\lambda_{\min}(\mathbf{P})}<\min\{r(\gamma_1),r_0\}$, where $r_0<\min_{i\in\mathsf{V}}\min\{|\mathbf{x}_i^*-p_i|,|\mathbf{x}_i^*-q_i|\}$. Note that γ_1 is chosen such that the derivative of V along (16) in the ball $\mathscr{B}(\mathbf{z}^*,r(\gamma_1))$ is nonpositive. Then, take $V_{\min}=\inf_{\|\hat{\mathbf{z}}\|=r(\gamma_1)}V(\hat{\mathbf{z}})$. One obtains $\{\hat{\mathbf{z}}:V(\hat{\mathbf{z}})\leq V_{\min}\}\subset\mathscr{B}(\mathbf{z}^*,r(\gamma_1))$. By letting $r\leq\frac{\min\{r(\gamma_1),r_0\}}{\sqrt{\lambda_{\max}(\mathbf{P})/\lambda_{\min}(\mathbf{P})}}$

$$\mathscr{B}(\mathbf{z}^*,r)\subset\{\hat{\mathbf{z}}:\,V(\hat{\mathbf{z}})\leq V_{\min}\}\subset\mathscr{B}(\mathbf{z}^*,r(\gamma_1)).$$

This implies that if $\|\hat{\mathbf{z}}(0)\|\leq r$, then $\dot{V}(t)\leq 0$ for all t. As a result, $\hat{\mathbf{z}}^\top(t)\mathbf{P}\hat{\mathbf{z}}(t)\leq r^2\lambda_{\max}(\mathbf{P})$ for all $t\geq 0$. This in turn yields that $\|\hat{\mathbf{z}}(t)\|\leq r\sqrt{\lambda_{\max}(\mathbf{P})/\lambda_{\min}(\mathbf{P})}$ and $\dot{V}(t)<0$ for all t. This, together with Lyapunov stability theory [36], gives that $\hat{\mathbf{z}}(t)\to\mathbf{0}$ as time goes to infinity, which completes the proof. ■

C. Proof of Theorem 2(iii)

We prove that local constraint consensus can be realized for system (1) with $\mathbf{H}_{ij}=\chi_j, i,j=1,\ldots,N$. Replacing χ_i for all i with the identity function, system (1) turns out to be $\dot{\mathbf{z}}=\Sigma\mathbf{z}$ with $\Sigma=\begin{bmatrix}\mathbf{0}_{N\times N} & \mathbf{I}_N\\ -\mathbf{L} & -k\mathbf{I}_N\end{bmatrix}$. By calculations for eigenvalues of Γ in Appendix C-B, if $k>\sqrt{(4\mathsf{Im}(\mu_i))^2/16\mathsf{Re}(\mu_i)}$, where μ_i is any nonzero eigenvalue of $\mathbf{L}$, then Σ has a single zero eigenvalue and all other eigenvalues have negative real parts. Moreover, $[\mathbf{1}_N^\top,\mathbf{0}_{1\times N}]^\top$ and $[\mathbf{w}^\top,\frac{1}{k}\mathbf{w}^\top]^\top$ are the right and left eigenvectors associated with eigenvalue zero, respectively. In addition, $\mathbf{w}_i$ is nonnegative [34] and $\mathbf{w}^\top\mathbf{1}_N=1$. According to Ren and Beard [4], if replacing χ_i for all i with an identity function, then $\mathbf{z}\to\mathbf{z}^*=[(\mathbf{x}^*)^\top,(\mathbf{v}^*)^\top]^\top$ (i.e., $\mathbf{x}_i\to\mathbf{w}^\top\mathbf{x}(0)+(1/k)\mathbf{w}^\top\mathbf{v}(0)=\mathbf{w}^\top\mathbf{x}(0)$ and $\mathbf{v}_i\to 0$) exponentially fast as $t\to\infty$. It is worth pointing out that $\mathbf{w}^\top\mathbf{x}(0)$ is a convex combination of $\mathbf{x}_i(0), i=1,\ldots,N$.

By recalling the exponential convergence of $\mathbf{z}(t)$ to $\mathbf{z}^*$, there holds that $|\mathbf{v}_i(t)|\leq\gamma_1e^{-\gamma_2t}\|\mathbf{z}(0)-\mathbf{z}^*\|=\gamma_1e^{-\gamma_2t}\|\mathbf{e}(0)\|$, where $\gamma_i>0, i=1,2$ are some constants and irrespective of the initial values, and $\mathbf{e}(0)=\mathbf{z}(0)-\mathbf{z}^*$. Therefore

$$\begin{aligned}\|\mathbf{x}_i(t)-\mathbf{x}_i^*\|&\leq\|\mathbf{x}_i(0)-\mathbf{x}_i^*\|+\left\|\int_0^t\gamma_1e^{-\gamma_2\tau}\mathrm{d}\tau\right\|\cdot\|\mathbf{e}(0)\|\\ &\leq\|\mathbf{z}(0)-\mathbf{z}^*\|+\|\mathbf{e}(0)\|\int_0^t\gamma_1e^{-\gamma_2\tau}\mathrm{d}\tau\\ &=\|\mathbf{e}(0)\|\left(1+\int_0^t\gamma_1e^{-\gamma_2\tau}\mathrm{d}\tau\right).\end{aligned}$$

Let $\mathscr{B}=[\hat{p},\hat{q}]$, where $\hat{p}>p_*=\max_{j\in\mathsf{V}}p_j$ and $\hat{q}<q_*=\min_{j\in\mathsf{V}}q_j$. If $\|\mathbf{e}(0)\|$ is chosen to satisfy that $\|\mathbf{e}(0)\|(1+\int_0^t\gamma_1e^{-\gamma_2\tau}\mathrm{d}\tau)<r_1$ for all t, where $r_1=\min\{|\hat{p}-p_*|,|\hat{q}-q_*|\}$, then $\mathbf{x}_i(t)\in[p_*,q_*]$ for all $t\geq 0$ because $\mathbf{x}_i^*$ is a convex combination of $\mathbf{x}_i(0)$ and $\mathbf{x}_i^*\in[\hat{p},\hat{q}]$. We have shown that if $\mathbf{x}_i(0)\in\mathscr{B}$, $\mathbf{v}_i(0)=0$, and $\delta=\|\mathbf{e}(0)\|$ is less than a certain threshold, then $\mathbf{z}(t)-\mathbf{z}^*\to\mathbf{0}$ as $t\to\infty$ and $\mathbf{x}_i(t)\in[p_*,q_*]$ for all i and all $t\geq 0$. Hence, local constraint consensus can be guaranteed, which completes the proof. ■

APPENDIX D
PROOF OF PROPOSITION 1

According to Theorem 1, an equilibrium point always exists. The proof is straightforward by the observation that any equilibrium point $\mathbf{z}^*$ satisfies that $\sum_{j\in\mathcal{N}_i}\alpha_i^jp_j\leq\mathbf{x}_i^*=\sum_{j\in\mathcal{N}_i}\alpha_i^j\chi_j(\mathbf{x}_j^*)\leq\sum_{j\in\mathcal{N}_i}\alpha_i^jq_j\quad\forall i\in\mathsf{V}$. ■

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APPENDIX E
PROOF OF THEOREM 3

First, according to Lemma 3, the evolution of $\mathbf{z}_i(t) = [\mathbf{x}_i(t), \mathbf{v}_i(t)]^\top$ is expressible as

$$\mathbf{z}_i(t) = e^{\mathbf{A}_i t}\mathbf{z}_i(0) + \underbrace{\int_0^t e^{\mathbf{A}_i(t-\tau)}\begin{bmatrix}0\\ s_i(\tau)\end{bmatrix}\mathrm{d}\tau}_{\mathbf{h}(t)=[\mathbf{h}_1(t),\mathbf{h}_2(t)]^\top},\quad t\geq 0$$

where $\mathbf{A}_i$ is Hurwitz and $s_i(t) = \sum_{j\in\mathcal{N}_i} a_{ij}\chi_j(\mathbf{x}_j)$. Since $\mathcal{I}_m = \mathcal{I}_n$ for any $m, n \in \mathsf{V}$, one has $d_i p_* \leq s_i(t) \leq d_i q_*$, where $p_* = p_i$ and $q_* = q_i$ for all i. Without loss of generality, let $p_* = -q_*$. That is to say, we assume that the center of the interval $\cap_{j=1}^N \mathcal{I}_j$ is zero. Actually, one can realize this by appropriate translational movement of the x-axis.

The eigenvalues of $\mathbf{A}_i$ are, respectively, $\lambda_1 = \frac{-k+\sqrt{k^2-4d_i}}{2}$ and $\lambda_2 = \frac{-k-\sqrt{k^2-4d_i}}{2}$. Via calculations, if $k^2 \neq 4d_i$, the associated eigenvectors are $\mathbf{w}_1 = [1, \lambda_1]^\top$ and $\mathbf{w}_2 = [1, \lambda_2]^\top$, respectively. Therefore, with $\mathbf{T} = [\mathbf{w}_1, \mathbf{w}_2]$, one has $\mathbf{T}^{-1}\mathbf{A}_i\mathbf{T} = \begin{bmatrix}\lambda_1 & \\ & \lambda_2\end{bmatrix}$, where $\mathbf{T}^{-1} = \begin{bmatrix}-\frac{\lambda_2}{\lambda_1-\lambda_2} & \frac{1}{\lambda_1-\lambda_2}\\ -\frac{\lambda_1}{\lambda_2-\lambda_1} & \frac{1}{\lambda_2-\lambda_1}\end{bmatrix}$. As a result, it can be obtained that

$$\begin{aligned}\begin{bmatrix}\mathbf{h}_1(t)\\ \mathbf{h}_2(t)\end{bmatrix} &= \int_0^t \exp\{\mathbf{A}_i(t-\tau)\}\begin{bmatrix}0\\ s_i(\tau)\end{bmatrix}\mathrm{d}\tau\\ &= \mathbf{T}\int_0^t \begin{bmatrix}e^{\lambda_1(t-\tau)} & \\ & e^{\lambda_2(t-\tau)}\end{bmatrix}\mathbf{T}^{-1}\begin{bmatrix}0\\ s_i(\tau)\end{bmatrix}\mathrm{d}\tau\\ &= \mathbf{T}\begin{bmatrix}\frac{1}{\lambda_1-\lambda_2} & \\ & \frac{1}{\lambda_2-\lambda_1}\end{bmatrix}\int_0^t \begin{bmatrix}e^{\lambda_1(t-\tau)} & \\ & e^{\lambda_2(t-\tau)}\end{bmatrix}\\ &\quad\times\begin{bmatrix}s_i(\tau)\\ s_i(\tau)\end{bmatrix}\mathrm{d}\tau\\ &\overset{\mathbf{T}=[\mathbf{w}_1,\mathbf{w}_2]}{=} \int_0^t \begin{bmatrix}s_i(\tau)\left[\frac{e^{\lambda_1(t-\tau)}}{\lambda_1-\lambda_2} - \frac{e^{\lambda_2(t-\tau)}}{\lambda_1-\lambda_2}\right]\\ \star\end{bmatrix}\mathrm{d}\tau.\end{aligned}$$

If $k^2 > 4d_i$, then λ_1 and λ_2 are real numbers and $\lambda_1 - \lambda_2 > 0$. As a consequence

$$\begin{aligned}\mathbf{h}_1(t) &= \int_0^t s_i(\tau)\left[\frac{e^{\lambda_1(t-\tau)}}{\lambda_1-\lambda_2} - \frac{e^{\lambda_2(t-\tau)}}{\lambda_1-\lambda_2}\right]\mathrm{d}\tau\\ &\leq d_i q_* \int_0^t \left[\frac{e^{\lambda_1(t-\tau)}}{\lambda_1-\lambda_2} - \frac{e^{\lambda_2(t-\tau)}}{\lambda_1-\lambda_2}\right]\mathrm{d}\tau\\ &\leq d_i q_*\left(\frac{1}{d_i} + \frac{\lambda_2 e^{\lambda_1 t} - \lambda_1 e^{\lambda_2 t}}{\lambda_1\lambda_2(\lambda_1-\lambda_2)}\right)\end{aligned}$$

which implies that $\limsup_{t\to\infty} \mathbf{h}_1(t) \leq q_*$. Similarly, one has $\liminf_{t\to\infty} \mathbf{h}_1(t) \geq p_*$.

As a result of the aforementioned discussions, $\mathbf{x}_i(t) \to [p_*, q_*]$ as $t \to \infty$, i.e., $\lim_{t\to\infty} \inf_{y\in[p_*,q_*]} |\mathbf{x}_i - y| = 0$ in an exponential manner if $k^2 > 4d_i$. In view of this, write system (1) into

$$\dot{\mathbf{z}} = \underbrace{\begin{bmatrix}\mathbf{0}_{N\times N} & \mathbf{I}_N\\ -\mathbf{L} & -k\mathbf{I}_N\end{bmatrix}}_{\Sigma}\mathbf{z} + \begin{bmatrix}\mathbf{0}\\ \mathbf{s}(t)\end{bmatrix} \tag{17}$$

where $\mathbf{s}(t) \in \mathbb{R}^N$ and its ith entry $\mathbf{s}_i(t) = \sum_{j=1}^N a_{ij}(\chi_j(\mathbf{x}_j) - \mathbf{x}_j)$ satisfies that $\mathbf{s}_i(t) \to 0$ as $t \to \infty$. Defining $\Delta = \begin{bmatrix}1 & \mathbf{0}_{N-1}^\top\\ \mathbf{1}_{N-1} & -\mathbf{I}_{N-1}\end{bmatrix}$, which satisfies $\Delta\Delta = \mathbf{I}$ and multiplying both sides of (17) with $\mathrm{diag}\{\Delta, \Delta\}$, one has

$$\dot{\hat{\mathbf{z}}} = \underbrace{\begin{bmatrix}\mathbf{0}_{(N-1)\times(N-1)} & \mathbf{I}_{N-1}\\ -\hat{\mathbf{L}} & -k\mathbf{I}_{N-1}\end{bmatrix}}_{\hat{\Sigma}}\hat{\mathbf{z}} + \underbrace{\begin{bmatrix}\mathbf{0}\\ \hat{\mathbf{s}}(t)\end{bmatrix}}_{\hat{\mathbf{b}}}$$

where $\hat{\mathbf{z}} = \mathrm{diag}\left\{\left[\mathbf{1}_{N-1}, -\mathbf{I}_{N-1}\right], \left[\mathbf{1}_{N-1}, -\mathbf{I}_{N-1}\right]\right\}\mathbf{z}$ and $\hat{\mathbf{L}} = \left[\mathbf{1}_{N-1}, -\mathbf{I}_{N-1}\right]\mathbf{L}\left[\mathbf{0}_{N-1}, -\mathbf{I}_{N-1}\right]^\top$ is a Hurwitz matrix because of strong connectivity of $\mathcal{G}$ [9]. Moreover, by the proof of Theorem 2, $\hat{\Sigma}$ is also a Hurwitz matrix if $k > \max_{i\in\{1,\ldots,N\}} |\mathsf{Im}(\mu_i)|/\sqrt{\mathsf{Re}(\mu_i)}$, where μ_i is an eigenvalue of $\hat{\mathbf{L}}$. In addition, $\hat{\mathbf{s}}(t) \to \mathbf{0}$ exponentially fast as $t \to \infty$. Note that global asymptotic convergence to state consensus for system (1) holds if and only if $\lim_{t\to\infty} \hat{\mathbf{z}}(t) = \mathbf{0}$ [9]. Now, if $k^2 > 4d_i$ for all i and $k > \max_{i\in\{1,\ldots,N\}} |\mathsf{Im}(\mu_i)|/\sqrt{\mathsf{Re}(\mu_i)}$, then we have $\|\exp\{\hat{\Sigma}t\}\| \leq \gamma_1 e^{-\gamma_2 t}$ [37] and $\|\hat{\mathbf{b}}(t)\| \leq \gamma_3 e^{-\gamma_4 t}$ for some $\gamma_i > 0, i = 1, \ldots, 4$. It is thus readily obtained that $\|\hat{\mathbf{z}}(t)\| \leq \gamma_1 e^{-\gamma_2 t}\|\hat{\mathbf{z}}(0)\| + \int_0^t \gamma_1 e^{-\gamma_2(t-\tau)}\gamma_3 e^{-\gamma_4\tau}\mathrm{d}\tau$, which gives that $\hat{\mathbf{z}}(t) \to \mathbf{0}$ as $t \to \infty$. Hence, state consensus for (1) can be achieved. Moreover, from system (1), for any i, there holds that $\mathbf{v}_i(t) = e^{-kt}\mathbf{v}_i(0) + \int_0^t e^{-k(t-\tau)}(\sum_{j\in\mathcal{N}_i} a_{ij}\left(\chi_j(\mathbf{x}_j) - \mathbf{x}_i\right))\,\mathrm{d}\tau$, which implies that $\mathbf{v}_i(t) \to 0$ since $\sum_{j\in\mathcal{N}_i} a_{ij}(\chi_j(\mathbf{x}_j) - \mathbf{x}_i) \to 0$ exponentially fast as $t \to \infty$ ■

APPENDIX F
PROOF OF THEOREM 4

(i) Consider the case $\cap_{j=1}^N = \emptyset$. Let $\mathbf{z}^* = [(\mathbf{x}^*)^\top, \mathbf{0}_{1\times N}]^\top$ be an equilibrium point. Then, the following system of nonlinear equations hold:

$$\sum_{j\in\mathcal{N}_i} a_{ij}\left(\chi_i\left(\mathbf{x}_j^*\right) - \mathbf{x}_i^*\right) = 0, i = 1, \ldots, N.$$

This in turn gives that $\mathbf{x}_i^* = \sum_{j\in\mathcal{N}_i} \alpha_i^j \chi_i(\mathbf{x}_j^*)$, where $\alpha_i^j = a_{ij}/\sum_{j\in\mathcal{N}_i} a_{ij}$. In view of the aforementioned fact, the existence and uniqueness of the solution to (2) can be proved by following the idea used to prove Theorem 1. We omit the lengthy details. In addition, $p_i \leq \chi_i(\mathbf{x}_j^*) \leq q_i$. Therefore, $\mathbf{x}_i^*$ must be within the interval $[p_i, q_i]$, that is $p_i \leq \mathbf{x}_i^* \leq q_i$.

(ii) Theorem 4(ii) can be proved by following the arguments developed to prove Lemma 4 and Corollary 1.

(iii) Since $\mathcal{I}_m \cap \mathcal{I}_n = \emptyset$ for any $m, n \in \mathsf{V}$, an equilibrium point exists and is unique by Theorem 4(i). Moreover, $p_i \leq \mathbf{x}_i^* \leq q_i$ for all i. Hence, $\chi_i(\mathbf{x}_j^*) = p_i$ or $\chi_i(\mathbf{x}_j^*) = q_i$ for any $j \in \mathcal{N}_i$.

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Similarly to Lemma 5, write the system of nonlinear equations (2) into the following linear compact form:

$$\mathbf{A}\mathbf{x}^* = \mathbf{b}, \quad s.t. \quad \mathbf{x}_i^* \in [p_i, q_i], \ i = 1, \ldots, N \tag{18}$$

where $\mathbf{A} = \text{diag}\{\sum_{j\in\mathcal{N}_1} a_{1j}, \ldots, \sum_{j\in\mathcal{N}_N} a_{Nj}\}$,$\mathbf{b} = \text{diag}\{\sum_{j\in\mathcal{N}_1} a_{1j}y_j^1, \ldots, \sum_{j\in\mathcal{N}_N} a_{Nj}y_j^N\}$, and $y_j^i = p_i$ if $q_j \le p_i$ or $y_j^i = q_i$ if $p_j \ge q_i$.

Since $\mathcal{G}$ is strongly connected, $\sum_{j\in\mathcal{N}_i} a_{ij}$ is nonzero for every $i \in \mathsf{V}$. Therefore, the matrix $\mathbf{A}$ is invertible. We first write $\mathbf{z}^* = \mathbf{A}^{-1}\mathbf{b}$. Next, we verify that the entries of $\mathbf{x}^*$ satisfy that $p_i \le \mathbf{x}_i^* \le q_i$. The inverse of $\mathbf{A}$ is $\mathbf{A}^{-1} = \text{diag}\{(\sum_{j\in\mathcal{N}_1} a_{1j})^{-1}, \ldots, (\sum_{j\in\mathcal{N}_N} a_{Nj})^{-1}\}$. Hence, the ith entry of $\mathbf{A}^{-1}\mathbf{b}$ is the convex combination of p_i and q_i for $i = 1, \ldots, N$. That is to say, $\mathbf{x}_i^* \in [p_i, q_i]$ for all i. ■

Appendix G
Proof of Theorem 5

The empty intersection of constraint intervals ensures that there exists a unique equilibrium point for system (1). Moreover, if the unique equilibrium point is also an E-IEP, then the right-hand side of (1) is differentiable at this equilibrium point. By Theorem 4, there exists $i \in \mathsf{V}$ such that $\chi_j(\mathbf{x}_i^*) \ne \mathbf{x}_i^*$ for some j. Then, for the neighbor-interval-consensus framework, it follows from the arguments developed in Lemma 5 that the system of nonlinear equations (2) can also be written as $-\hat{\mathbf{L}}_2\mathbf{x}^* + \mathbf{b} = \mathbf{0}$, where $-\hat{\mathbf{L}}_2$ has eigenvalues with positive real parts and $\mathbf{b}$ is some constant.

By invoking Lyapunov's indirect method, one can obtain a Jacobian matrix of system (1) at the E-IEP: $\begin{bmatrix} \mathbf{0}_{N\times N} & \mathbf{I}_N \\ -\hat{\mathbf{L}}_2 & -k\mathbf{I}_N \end{bmatrix}$. Note that this matrix determines the local stability of the E-IEP. The remaining proofs for Theorem 5(i) and (ii) can be completed by mimicking the proofs for Theorem 2(i) and (ii); while the result for the nonempty intersection, as is stated in Theorem 5(iii), can be proved by mimicking the proof for Theorem 2(iii). The lengthy details are omitted here for space limitation.

■

Appendix H
Proof of Proposition 2

Since $\cap_{j=1}^N I_j = \emptyset$, according to Theorem 4, an equilibrium point always exists. The proof is straightforward by the observation that any equilibrium point $\mathbf{z}^*$ satisfies that $p_i \le \mathbf{x}_i^* = \sum_{j\in\mathcal{N}_i} \alpha_i^j \chi_i(\mathbf{x}_j^*) \le q_i \quad \forall i \in \mathsf{V}$. Here, $\alpha_i^j = \frac{a_{ij}}{\sum_{j\in\mathcal{N}_i} a_{ij}}$. ■

Appendix I
Proof of Theorem 6

From Theorem 4, there exists a unique equilibrium point $\mathbf{z}^*$. Let $\mathbf{z}_i(0)$ be given arbitrarily. Recall that $\mathbf{z}_i(t) = e^{\mathbf{A}_i t}\mathbf{z}_i(0) + \int_0^t e^{\mathbf{A}_i(t-\tau)} \begin{bmatrix} 0 \\ s_i(\tau) \end{bmatrix} \mathrm{d}\tau$, where $d_i p_i \le s_i(t) = \sum_{j\in\mathcal{N}_i} a_{ij}(\chi_i(\mathbf{x}_j) \le d_i q_i$.

Similar to the proof of Theorem 3, assume without loss of generality that the center of interval $\mathcal{I}_i = [p_i, q_i]$ is zero, i.e., $p_i = -q_i$. Applying the arguments developed in the proof of Theorem 3 to the constraint interval $[p_i, q_i]$ instead of $[p_*, q_*]$, one has that $\mathbf{x}_i(t)$ approaches $[p_i, q_i]$ exponentially fast as $t \to \infty$, i.e., $\lim_{t\to\infty} \inf_{y\in[p_i,q_i]} |\mathbf{x}_i - y| = 0$. Since given any $j \in \mathsf{V}$, $\mathcal{I}_j \cap \mathcal{I}_k = \emptyset$ for all k, there exists a $T > 0$ such that $\mathbf{x}_k(t) \notin \mathcal{I}_j$, $t \ge T$, for all $j \ne k$ and, hence, $s_i(t) = \hat{s}_i \in [d_i p_i, d_i q_i]$, $t \ge T$, for all i. As a consequence, for $t \ge T$, there holds that $\mathbf{z}_i(t) = e^{\mathbf{A}_i(t-T)}\mathbf{z}_i(T) + \int_T^t e^{\mathbf{A}_i(t-\tau)} \begin{bmatrix} 0 \\ \hat{s}_i \end{bmatrix} \mathrm{d}\tau$, $t \ge T$, where by Lemma 3, $\mathbf{z}_i(T)$ is bounded. A simple computation yields $\mathbf{x}_i(t) \to \mathbf{x}_i^* = \hat{s}_i/d_i \in [p_i, q_i]$ and $\mathbf{v}_i(t) \to 0$ as $t \to \infty$, for which the details are omitted for brevity. ■

References

[1] L. Moreau, "Stability of multiagent systems with time-dependent communication links," *IEEE Trans. Autom. Control*, vol. 50, no. 2, pp. 169–182, Feb. 2005.

[2] M. Cao, A. S. Morse, and B. D. Anderson, "Reaching a consensus in a dynamically changing environment: Convergence rates, measurement delays, and asynchronous events," *SIAM J. Control Optim.*, vol. 47, no. 2, pp. 601–623, 2008.

[3] A. V. Proskurnikov and R. Tempo, "A tutorial on modeling and analysis of dynamic social networks Part I," *Annu. Rev. Control*, vol. 43, pp. 65–79, 2017.

[4] W. Ren and R. W. Beard, *Distributed Consensus in Multi-Vehicle Cooperative Control*. Berlin, Germany: Springer, 2008.

[5] L. Wang and F. Xiao, "Finite-time consensus problems for networks of dynamic agents," *IEEE Trans. Autom. Control*, vol. 55, no. 4, pp. 950–955, Apr. 2010.

[6] V. Amelkin, F. Bullo, and A. K. Singh, "Polar opinion dynamics in social networks," *IEEE Trans. Autom. Control*, vol. 62, no. 11, pp. 5650–5665, Nov. 2017.

[7] A. Nedic, A. Ozdaglar, and P. A. Parrilo, "Constrained consensus and optimization in multi-agent networks," *IEEE Trans. Autom. Control*, vol. 55, no. 4, pp. 922–938, Apr. 2010.

[8] C. Altafini, "Consensus problems on networks with antagonistic interactions," *IEEE Trans. Autom. Control*, vol. 58, no. 4, pp. 935–946, Apr. 2013.

[9] J. Qin, Q. Ma, W. X. Zheng, H. Gao, and Y. Kang, "Robust H_∞ group consensus for interacting clusters of integrator agents," *IEEE Trans. Autom. Control*, vol. 62, no. 7, pp. 3559–3566, Jul. 2017.

[10] V. D. Blondel, J. M. Hendrickx, and J. N. Tsitsiklis, "On Krause's multi-agent consensus model with state-dependent connectivity," *IEEE Trans. Autom. Control*, vol. 54, no. 11, pp. 2586–2597, Nov. 2009.

[11] L. Ding, W. X. Zheng, and G. Guo, "Network-based practical set consensus of multi-agent systems subject to input saturation," *Automatica*, vol. 89, pp. 316–324, 2018.

[12] J. Lorenz, "Continuous opinion dynamics under bounded confidence: A survey," *Int. J. Modern Phys. C*, vol. 18, no. 12, pp. 1819–1838, 2007.

[13] F. Ceragioli, G. Lindmark, C. Veibäck, N. Wahlström, M. Lindfors, and C. Altafini, "A bounded confidence model that preserves the signs of the opinions," in *Proc. Eur. Control Conf.*, Aalborg, Denmark, 2016, pp. 543–548.

[14] V. D. Blondel, J. M. Hendrickx, and J. N. Tsitsiklis, "Continuous-time average-preserving opinion dynamics with opinion-dependent communications," *SIAM J. Control Optim.*, vol. 48, no. 8, pp. 5214–5240, 2010.

[15] F. Dietrich, S. Martin, and M. Jungers, "Control via leadership of opinion dynamics with state and time-dependent interactions," *IEEE Trans. Autom. Control*, vol. 63, no. 4, pp. 1200–1207, Apr. 2018.

[16] A. Li, S. P. Cornelius, Y.-Y. Liu, L. Wang, and A.-L. Barabási, "The fundamental advantages of temporal networks," *Science*, vol. 358, no. 6366, pp. 1042–1046, 2017.

[17] A. Fontan, G. Shi, X. Hu, and C. Altafini, "Interval consensus: A novel class of constrained consensus problems for multiagent networks," in *Proc. 56th IEEE Conf. Decis. Control*, Melbourne, VIC, Australia, 2017, pp. 4155–4160.

[18] A. Fontan, G. Shi, X. Hu, and C. Altafini, "Interval consensus for multiagent networks," *IEEE Trans. Autom. Control*, vol. 65, no. 5, pp. 1855–1869, May 2020.

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[19] G. Binetti, A. Davoudi, D. Naso, B. Turchiano, and F. L. Lewis, "A distributed auction-based algorithm for the nonconvex economic dispatch problem," *IEEE Trans. Ind. Inform.*, vol. 10, no. 2, pp. 1124–1132, May 2014.
[20] J. Ghaderi and R.Srikant, "Opinion dynamics in social networks with stubborn agents: Equilibrium and convergence rate," *Automatica*, vol. 50, no. 12, pp. 3209–3215, 2014.
[21] D. Acemoğlu, G. Como, F. Fagnani, and A. Ozdaglar, "Opinion fluctuations and disagreement in social networks," *Math. Oper. Res.*, vol. 38, no. 1, pp. 1–27, 2013.
[22] E. Yildiz, A. Ozdaglar, D. Acemoglu, A. Saberi, and A. Scaglione, "Binary opinion dynamics with stubborn agents," *ACM Trans. Econ. Comput.*, vol. 1, no. 4, pp. 1–30, 2013.
[23] A. V. Proskurnikov, R. Tempo, M. Cao, and N. E. Friedkin, "Opinion evolution in time-varying social influence networks with prejudiced agents," *IFAC-PapersOnLine*, vol. 50, no. 1, pp. 11 896–11 901, 2017.
[24] M. Ye, Y. Qin, A. Govaert, B. D. Anderson, and M. Cao, "An influence network model to study discrepancies in expressed and private opinions," *Automatica*, vol. 107, pp. 371–381, 2019.
[25] A. Fontan and C. Altafini, "Multiequilibria analysis for a class of collective decision-making networked systems," *IEEE Control Netw. Syst.*, vol. 5, no. 4, pp. 1931–1940, Dec. 2018.
[26] Z.-X. Liu and Z.-Q. Chen, "Discarded consensus of network of agents with state constraint," *IEEE Trans. Autom. Control*, vol. 57, no. 11, pp. 2869–2874, Nov. 2012.
[27] U. Lee and M. Mesbahi, "Constrained consensus via logarithmic barrier functions," in *Proc. 50th IEEE Conf. Decis. Control Eur. Control Conf.*, Orlando, FL, USA, 2011, pp. 3608–3613.
[28] M. Barić and F. Borrelli, "Distributed averaging with flow constraints," in *Proc. Amer. Control Conf.*, San Francisco, CA, USA, 2011, pp. 4834–4839.
[29] W. Meng, Q. Yang, J. Si, and Y. Sun, "Consensus control of nonlinear multiagent systems with time-varying state constraints," *IEEE Trans. Cybern.*, vol. 47, no. 8, pp. 2110–2120, Aug. 2017.
[30] S. Lee and A. Nedic, "Distributed random projection algorithm for convex optimization," *IEEE J. Sel. Topics Signal Process.*, vol. 7, no. 2, pp. 221–229, Apr. 2013.
[31] R. Bailo, M. Bongini, J. A. Carrillo, and D. Kalise, "Optimal consensus control of the Cucker-Smale model," *IFAC-PapersOnLine*, vol. 51, no. 13, pp. 1–6, 2018.
[32] P. M. DeMarzo, D. Vayanos, and J. Zwiebel, "Persuasion bias, social influence, and unidimensional opinions," *Quart. J. Econ.*, vol. 118, no. 3, pp. 909–968, 2003.
[33] A. Kappes, A. H. Harvey, T. Lohrenz, P. R. Montague, and T. Sharot, "Confirmation bias in the utilization of others' opinion strength," *Nat. Neurosci.*, vol. 23, no. 1, pp. 130–137, 2020.
[34] C. Godsil and G. F. Royle, *Algebraic Graph Theory*, vol. 207. Berlin, Germany: Springer, 2013.
[35] W. C. Wah, *Synchronization in Complex Networks of Nonlinear Dynamical Systems*. Singapore: World Scientific, 2007.
[36] H. K. Khalil and J. W. Grizzle, *Nonlinear Systems*, vol. 3. Upper Saddle River, NJ, USA: Prentice-Hall, 2002.
[37] R. A. Horn and C. R. Johnson, *Matrix Analysis*. Cambridge, U.K.: Cambridge Univ. Press, 2012.
[38] A. Aydogdu, M. Caponigro, S. McQuade, B. Piccoli, N. P. Duteil, F. Rossi, and E. Trelat, "Interaction network, state space and control in social dynamics," *Active Particles*, vol. 1, pp. 99–140, 2017.
[39] H. Luo, "Accelerated differential inclusion for convex optimization," 2021, *arXiv:2103.06629*.



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